

0 2 **Table 1** shows the nucleon number A and the nuclear radii R for three nuclides.

Table 1

Nuclide	A	$R / 10^{-15} \text{ m}$
Carbon	12	2.66
Iron	56	4.35
Lead	208	6.66

0 2 . 1 It is suggested that:

$$A = \frac{R^3}{k}$$

where k is a constant.

Use the data in **Table 1** to determine a reliable value for k .

[3 marks]

$$k = \text{_____} \text{ m}^3$$



0 2 . 2

Determine the percentage uncertainty in the value of k that you found in question **02.1**.

[1 mark]

percentage uncertainty in k = _____

0 2 . 3

The mass of a nucleon is 1.67×10^{-27} kg.

Show that the data in **Table 1** support the idea that nuclear material has a constant density.

[2 marks]

Question 2 continues on the next page

Turn over ►



0 2 . 4

This question is about Rutherford's scattering experiment.

The nuclear radius cannot be bigger than the distance of closest approach of an alpha particle.

Complete **Figure 2** to show the paths followed by each alpha particle.

The meaning of the symbols used in **Figure 2**:

→ indicates the initial direction of alpha particle

○ indicates target nucleus

[3 marks]

Figure 2



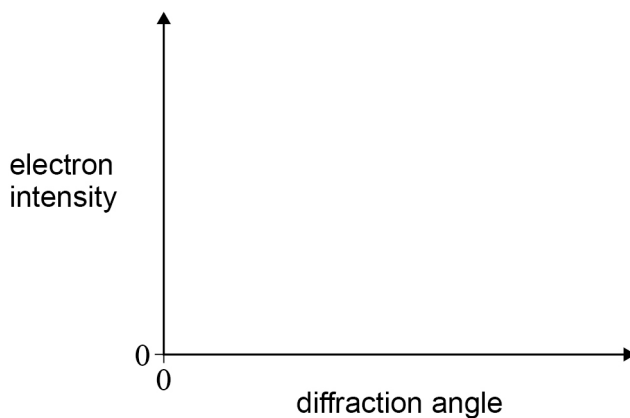
0 2 . 5

Nuclear radii can be found using the diffraction of high-energy electrons.

Sketch, on the axes below, a graph showing the variation of electron intensity with diffraction angle.

Label the angle θ used in the calculation of nuclear radius.

[2 marks]



Don't write

2	0
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The radius of a hydrogen nucleus is of the order of

[1 mark]

A 10^{-10} m.



B 10^{-12} m.



C 10^{-15} m.



D 10^{-18} m.



Turn over ►



Do not write

2

1

Cobalt-60 decays by β^- emission to form nickel-60.

What is true about the magnitude of the binding energy per nucleon of the nickel nuclide and the additional particle emitted in the decay?

[1 mark]

	Magnitude of the binding energy per nucleon of nickel-60	Additional particle emitted in the decay	
A	Less than the binding energy per nucleon of cobalt-60	Antineutrino	☐
B	Less than the binding energy per nucleon of cobalt-60	Neutrino	☐
C	Greater than the binding energy per nucleon of cobalt-60	Antineutrino	☐
D	Greater than the binding energy per nucleon of cobalt-60	Neutrino	☐

The binding energy of a nucleus is equivalent to

[1 mark]

- A** the energy released when the nucleus de-excites and emits a gamma ray.
- B** the sum of the kinetic energy of an emitted alpha particle and the recoil energy of the nucleus.
- C** the total energy required to assemble all of the particles in the nucleus.
- D** the total energy required to separate all of the particles in the nucleus.

☐☐☐☐

Don't write

Which row gives the binding energy of an iron-56 nucleus and of a proton?

[1 mark]

	Binding energy of an iron-56 nucleus / MeV	Binding energy of a proton / MeV
A	8.8	0
B	8.8	1
C	493	0
D	493	1

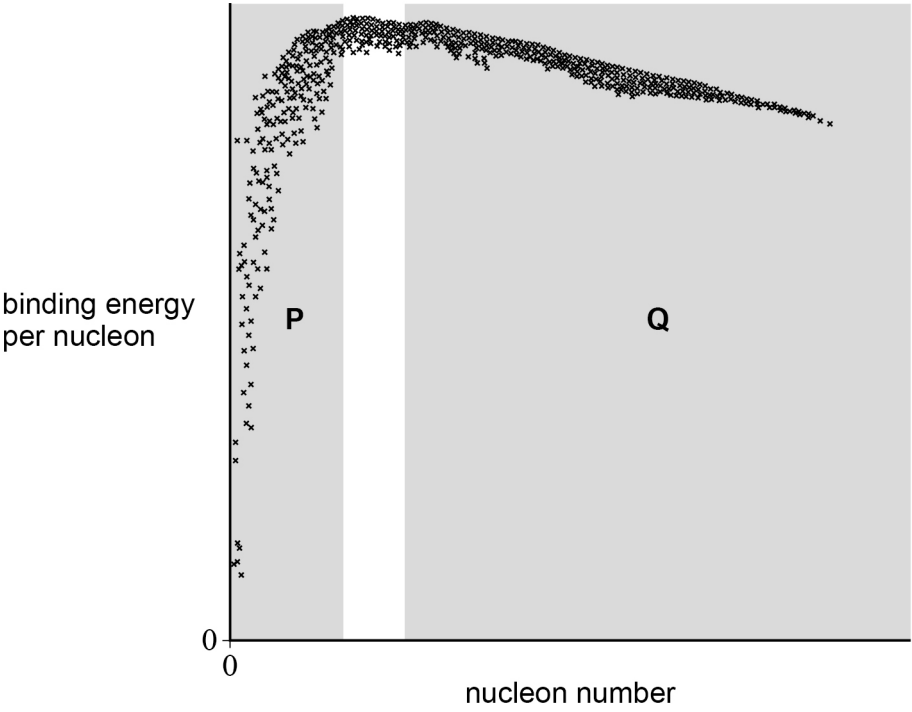
☐☐☐☐

Turn over for the next question



Do not write

The diagram shows a plot of binding energy per nucleon against nucleon number for the whole range of nuclides.



Which row shows regions of the plot in which nuclei can undergo fission and / or fusion accompanied by a net release of energy?

[1 mark]

	Fission	Fusion	
A	P	Q	☐
B	Q	P	☐
C	P and Q	P	☐
D	P and Q	P and Q	☐



Don't write

2 **5** The atomic mass unit u is defined as:

[1 mark]

A the rest mass of a free neutron.

☐

B the rest mass of a free proton.

☐

C $\frac{1}{12}$ of the mass of a carbon-12 atom.

☐

D $\frac{1}{16}$ of the mass of an oxygen-16 atom.

☐

Do not write
outside the
box

2**6**

The mass of an alpha particle is 6.644×10^{-27} kg.
A mass of 1.0×10^{-27} kg is equivalent to an energy of 561 MeV.

What is the binding energy of an alpha particle?

[1 mark]

A 27 MeV

☐

B 29 MeV

☐

C 84 MeV

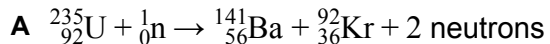
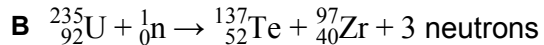
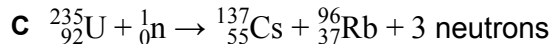
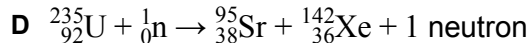
☐

D 3700 MeV

☐

Which is a possible thermal fission reaction?

[1 mark]

☐☐☐☐

Turn over ►



Do not write

2**8**

Do not write
outside the
box

In a nuclear fission reactor, it is useful to have a moderator that slows the neutrons down with the smallest possible number of collisions.

Which material would slow neutrons with the smallest number of collisions?

[1 mark]

A Hydrogen

☐

B Deuterium

☐

C Helium

☐

D Carbon

☐

Which row shows an appropriate choice of materials for the moderator, control rods and coolant for a thermal nuclear reactor?

[1 mark]

	Moderator	Control rods	Coolant	
A	Graphite	Uranium	Carbon dioxide	☐
B	Sodium	Boron	Carbon dioxide	☐
C	Water	Boron	Water	☐
D	Water	Uranium	Water	☐

3**0**

What is the order of magnitude of the temperature at the centre of the Sun?

[1 mark]

A 10^3 K

☐

B 10^5 K

☐

C 10^7 K

☐

D 10^9 K

☐

Don't write

3**1***Do not write
outside the
box*

Which is **not** a problem with the containment and shielding of a nuclear fusion reactor?

[1 mark]

A The high temperature of the plasma would cause the shielding material to melt.

☐

B Large amounts of radioactive beta emitters are produced in the fusion reactions.

☐

C Contact with a containment vessel would reduce the temperature of the plasma.

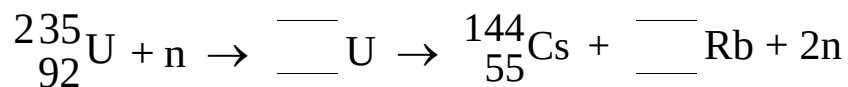
☐

D Large numbers of high-energy neutrons are produced in the fusion reactions.

☐

05

The two stages of an induced fission reaction are shown below.



05

1

Complete the equation.

[2 marks]

05

2

Table 1 shows the masses of the reactants and products of the reaction.

Table 1

	mass / 10^{-27} kg
${}_{92}^{235}\text{U}$	390.40797
n	1.67493
${}_{55}^{144}\text{Cs}$	239.07118
Rb	149.34849



Calculate the energy released during this fission reaction.

[3 marks]

energy released = _____ J

0 5 . 3 A neutron produced in this fission reaction has a kinetic energy of 2.0 MeV.

Show that the speed of the neutron is approximately $2 \times 10^7 \text{ m s}^{-1}$.

[2 marks]

Question 5 continues on the next page

Turn over ►



0 5 . 4

In one nuclear reactor, the moderator is heavy water that contains deuterium $\left({}_1^2\text{H}\right)$ nuclei.

The neutron in Question **05.3** has a head-on collision with a stationary deuterium nucleus. After the collision, the deuterium nucleus has a speed of $1.31 \times 10^7 \text{ m s}^{-1}$.

Determine the speed of the neutron immediately after the collision.

mass of a deuterium nucleus = $3.34 \times 10^{-27} \text{ kg}$

[2 marks]

speed = _____ m s^{-1}

0 5 . 5

Explain why there is a range of neutron speeds after neutrons have had one collision with a deuterium nucleus.

[2 marks]

END OF SECTION A

Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

2**1**

An alpha particle is directed at a target nucleus.
The initial velocity of the alpha particle is v .

The distance of closest approach of the alpha particle to the nucleus is

[1 mark]

A directly proportional to $\sqrt{v}$.

☐

B directly proportional to v^2 .

☐

C inversely proportional to $\sqrt{v}$.

☐

D inversely proportional to v^2 .

☐

2	2
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What is $\frac{\text{radius of } {}^{56}_{26}\text{Fe nucleus}}{\text{radius of } {}^{16}_8\text{O nucleus}}$?

[1 mark]

A 1.48

☐

B 1.52

☐

C 1.80

☐

D 1.87

☐

Turn over ►



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on this sheet

2 **3** The nuclear radius of ${}_{11}^{23}\text{Na}$ is 3.6 fm.

What is the density of the nucleus of ${}_{7}^{14}\text{N}$?

[1 mark]

A $9.4 \times 10^7 \text{ kg m}^{-3}$

☐

B $2.0 \times 10^8 \text{ kg m}^{-3}$

☐

C $9.4 \times 10^{16} \text{ kg m}^{-3}$

☐

D $2.0 \times 10^{17} \text{ kg m}^{-3}$

☐

Don't write

2**5**

What is the equivalent of 1 meV?

[1 mark]

A $1.1 \times 10^3 \text{ u}$

☐

B $1.1 \times 10^{-3} \text{ u}$

☐

C $1.1 \times 10^{-12} \text{ u}$

☐

D $1.1 \times 10^{-16} \text{ u}$

☐

Do not write
outside the
box

The equation shows a fission reaction for uranium-235.



Which shows the positive value for the change in mass for the reaction?

m_{U} = mass of U

m_{n} = mass of n

m_{B} = mass of Ba

m_{K} = mass of Kr

[1 mark]

A $m_{\text{B}} + m_{\text{K}} - m_{\text{U}} - 3m_{\text{n}}$

☐

B $m_{\text{B}} + m_{\text{K}} - m_{\text{U}} - 2m_{\text{n}}$

☐

C $m_{\text{U}} - m_{\text{B}} - m_{\text{K}} - 2m_{\text{n}}$

☐

D $m_{\text{U}} - m_{\text{B}} - m_{\text{K}} - 3m_{\text{n}}$

☐

Don't write

$^{239}_{94}\text{Pu}$ undergoes fission to form $^{134}_{54}\text{Xe}$ and $^{103}_{40}\text{Zr}$.

What is the energy released in this fission reaction?

binding energy per nucleon of $^{239}_{94}\text{Pu} = 7.56 \text{ MeV}$

binding energy per nucleon of $^{134}_{54}\text{Xe} = 8.41 \text{ MeV}$

binding energy per nucleon of $^{103}_{40}\text{Zr} = 8.43 \text{ MeV}$

[1 mark]

A 86 MeV

☐

B 107 MeV

☐

C 188 MeV

☐

D 2200 MeV

☐

Turn over for the next question



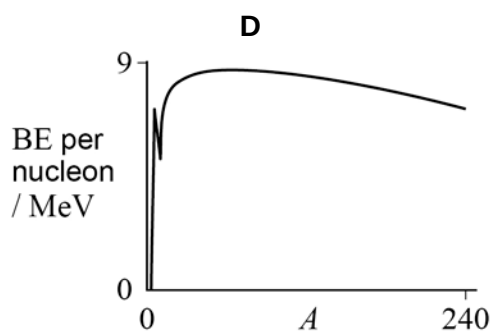
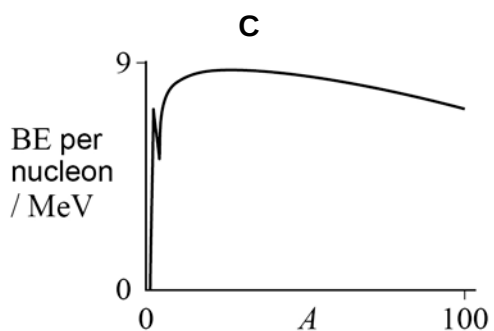
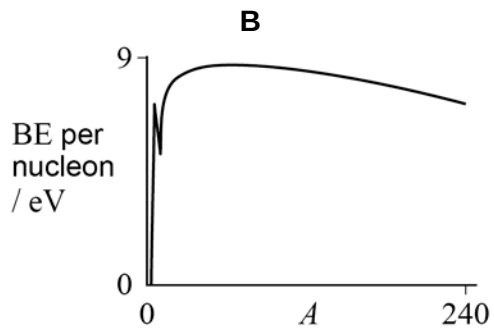
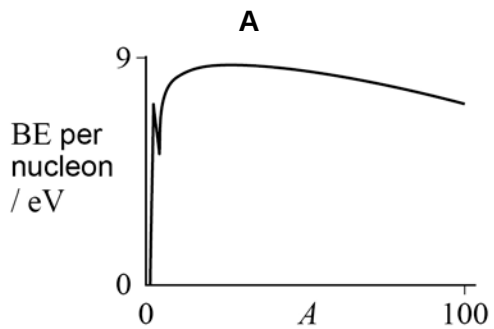
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2 8

Which is the most accurate representation of a plot of binding energy (BE) per nucleon against nucleon number A ?

[1 mark]

A ☐B ☐C ☐D ☐

Don't write

2

9

These statements are about mass changes in fusion and induced fission reactions in which there is a net release of energy.

Which statement is correct?

[1 mark]

A In induced fission, the total mass of the fission products is less than the mass of the fissile nucleus.

☐

B In induced fission, the total mass of the fission products is greater than the total mass of the reactants.

☐

C In fusion, the total mass of the products is less than the total mass of the reactants.

☐

D In fusion, the total mass of the products is greater than the total mass of the reactants.

☐

Which creates a significant risk from ionising radiation in the operation of a nuclear fusion reactor?

[1 mark]

A beta emitters produced by neutron absorption

☐

B the nuclei produced in the fusion reaction

☐

C alpha particles emitted in fusion

☐

D beta particles emitted in fusion

☐

Turn over for the next question

Turn over ►



Do not write

Do not write

0	2
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The mass of a nucleus of californium-252 ($^{252}_{98}\text{Cf}$) is 252.03 u.

0	2
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.	1
---	---

Calculate, in kg, the mass defect of a nucleus of californium-252.

[3 marks]

mass defect = _____ kg



0 2 . 2

Spontaneous fission of a californium-252 nucleus produces a xenon-140 ($^{140}_{54}\text{Xe}$) nucleus, a ruthenium-108 ($^{108}_{44}\text{Ru}$) nucleus and four neutrons.

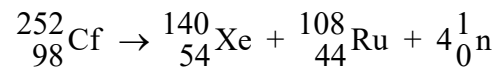


Table 1 gives the mass of each nuclide.

Table 1

Nuclide	Mass / u
$^{252}_{98}\text{Cf}$	252.03
$^{140}_{54}\text{Xe}$	139.89
$^{108}_{44}\text{Ru}$	107.88

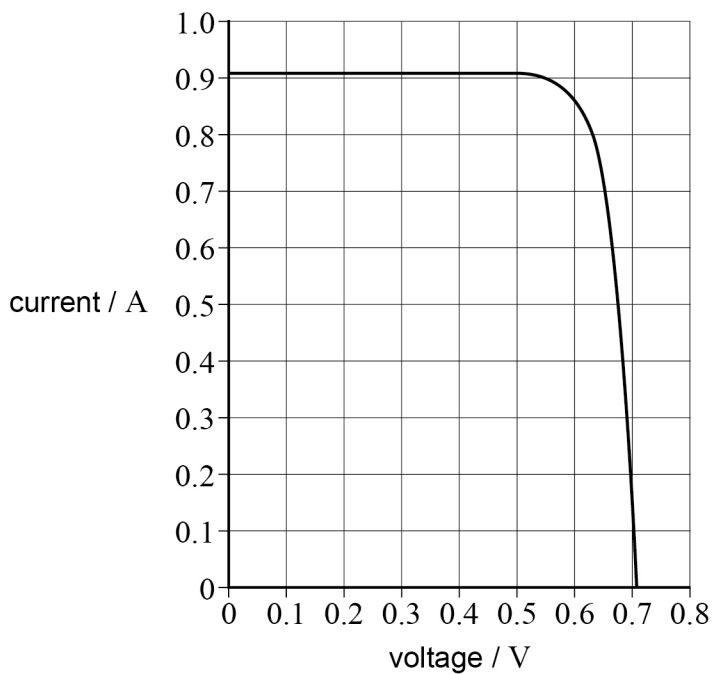
Calculate, in MeV, the amount of energy released in this fission.

[3 marks]

energy released = _____ MeV

0 3 . 1 Figure 1 shows the current–voltage characteristic for a solar cell.

Figure 1



Estimate the maximum power available from the solar cell.

[2 marks]

maximum power = _____ W



03.2

A different type of solar cell has an emf of 0.72 V and an internal resistance of $2.2\ \Omega$ when in use.

An array of solar cells of this type has an emf of 3.6 V and an internal resistance of $5.5\ \Omega$.

Deduce the number and arrangement of solar cells used to make this array.

[4 marks]

6

Turn over for the next question

Turn over ►



0 4

A gas has a density of 12 kg m^{-3} when its temperature is 860 K and its pressure is $2.1 \times 10^7 \text{ Pa}$.

0 4

1

Show that 60 kg of the gas contains approximately 9×10^{27} particles.

[3 marks]

0 4

2

Calculate the molar mass of the gas.

State an appropriate unit for your answer.

[3 marks]

molar mass = _____

unit = _____



0 4 . 3

The gas is heated at a rate of 75 MW as it flows through a pipe.
A mass of 60 kg of gas flows through the pipe each second.
The gas enters the pipe at a temperature of 860 K.

Calculate the temperature of the gas as it leaves the pipe.

specific heat capacity of the gas = $4.9 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

[3 marks]

temperature of the gas = _____ K

0 4 . 4

The hot gas now enters a cool turbine, forcing the turbine to rotate.

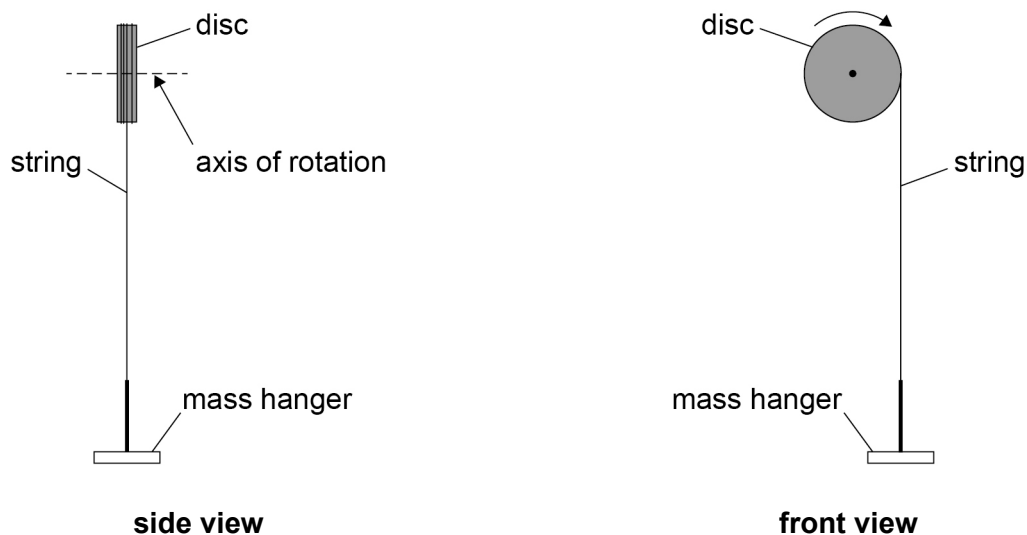
Discuss how the first law of thermodynamics applies to the gas as it forces the turbine to rotate.

[3 marks]



0 5

Figure 2 shows the equipment a student uses to find the moment of inertia of a disc. The student wraps a string of negligible mass several times around the disc. The student attaches a mass hanger to the free end of the string. The hanger is released from rest and falls. The disc rotates with a constant angular acceleration.

Figure 2

0 5 . 1

The disc makes exactly three rotations in a time of 2.2 s.

Show that the angular acceleration α of the disc is approximately 8 rad s^{-2} .

[3 marks]

The tension in the string exerts a torque on the disc.
The hanger has a mass m . It falls with a linear acceleration A .

0 5 . 2 Show that, in the absence of frictional forces,

$$\text{torque on the disc} = m(g - A)r$$

where r is the radius of the disc.

[2 marks]

0 5 . 3 The linear acceleration A is related to the angular acceleration α by

$$A = \alpha r$$

Determine, using the experimental data, the moment of inertia of the disc.

$$r = 2.0 \text{ cm}$$

$$m = 0.10 \text{ kg}$$

[3 marks]

moment of inertia = _____ kg m²

Question 5 continues on the next page

Turn over ►



0 5 . 4

In practice, frictional forces act on the disc as it rotates.

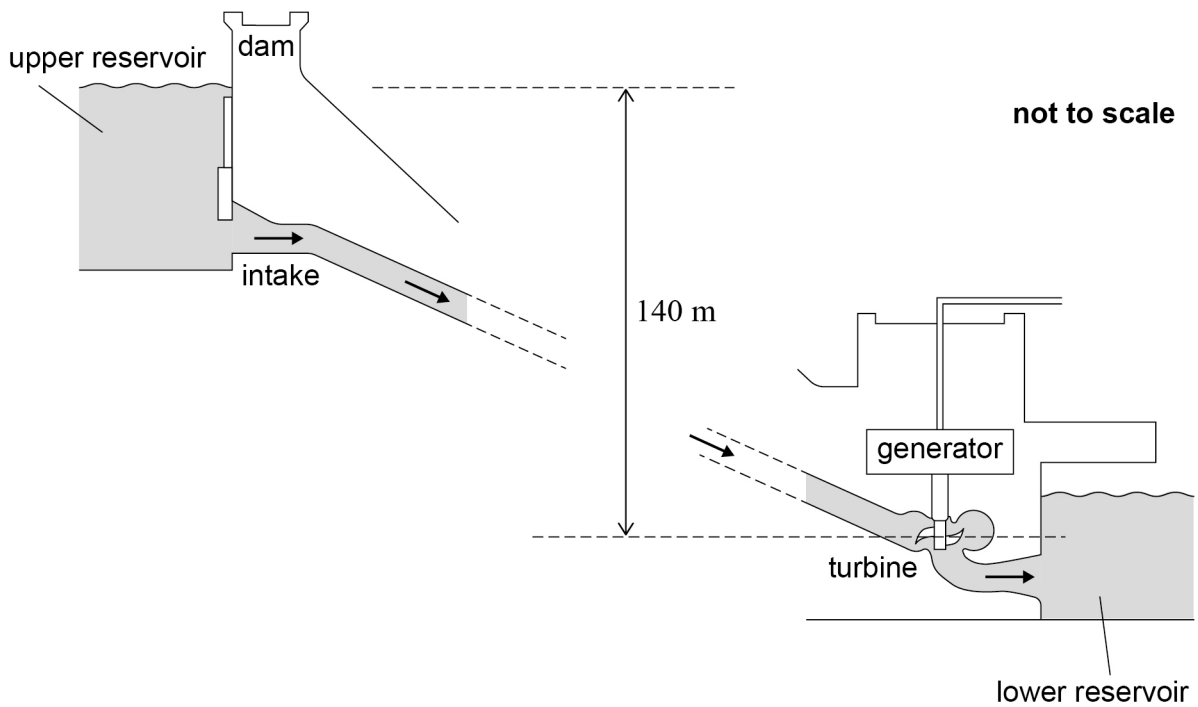
These frictional forces affect the accuracy of the experimental value for the moment of inertia of the disc.

Discuss why.

[3 marks]

Figure 3 shows a pumped storage system (PSS). Electricity is generated when water flows through a pipe from the upper reservoir to drive a turbine and a generator.

Figure 3



Question 6 continues on the next page

The water level in the upper reservoir is initially 140 m above the turbine.
The PSS generates electrical energy for an operating time T .
During this operating time:

- the water level decreases by 20 m
- 30 TJ of gravitational potential energy is transferred
- there is a constant electrical power output of 1.8 GW.

0 6 . 1

Calculate the mass of water that leaves the upper reservoir when the water level decreases by 20 m.

[2 marks]

mass = _____ kg

0 6 . 2

The efficiency of the transfer from gravitational potential energy to electrical energy is 82%.

Calculate T in hours.

[3 marks]

T = _____ hours



0 6 . 3

Outline why, during one complete cycle of operation, the efficiency of the PSS is less than 82%.

[2 marks]

0 6 . 4

Describe the benefits of using a PSS in an electrical power network.

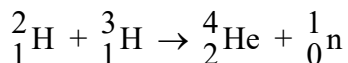
[3 marks]



07

Some proposed fusion reactors will use a fuel of deuterium $\left({}_1^2\text{H}\right)$ and tritium $\left({}_1^3\text{H}\right)$.

The fusion reaction produces helium-4 $\left({}_2^4\text{He}\right)$ and a neutron.



07.1

Explain why this reaction will occur only when the fuel is at a very high temperature.

[2 marks]

07.2

The temperature of the fuel in one reactor is 1.2 GK.

Calculate the average kinetic energy of a deuterium nucleus in the fuel.
Assume that the fuel behaves as an ideal gas.

[1 mark]

average kinetic energy = _____ J

07.3

State how the average kinetic energy of a **tritium** nucleus in the fuel compares to your answer to Question **07.2**.

[1 mark]



0 7 . 4

An estimate of the closest approach between a deuterium nucleus and a tritium nucleus can be made using the answers to Question **07.2** and Question **07.3**.

Explain how this estimate can be made.

[3 marks]

Question 7 continues on the next page

Turn over ►



0 7 . 5 **Table 2** shows the binding energy per nucleon for two nuclides involved in the fusion reaction.

The energy released in this fusion reaction is 17.589 MeV.

Table 2

Nuclide	Binding energy per nucleon / MeV
${}^2_1\text{H}$	1.1123
${}^4_2\text{He}$	7.0739

Calculate, in J, the binding energy of the tritium $\left({}^3_1\text{H}\right)$ nuclide.

[3 marks]

binding energy = _____ J



0 7 . 6

The fusion of a deuterium nucleus and a tritium nucleus initially produces a helium-5 nucleus. This nucleus quickly decays into an unexcited helium-4 nucleus and a neutron.

Show that the kinetic energy of the neutron is four times greater than the kinetic energy of the helium-4 nucleus.

Assume that the helium-5 nucleus is stationary before it decays.

[3 marks]

0 7 . 7

Designers of fusion reactors have to consider the risks associated with neutrons which have high kinetic energies.

Suggest **one** of these risks.

[1 mark]

14

END OF SECTION A

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

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CORRECT METHOD



WRONG METHODS



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Do **not** use additional sheets for this working.

0 8 At absolute zero, an ideal gas will

[1 mark]

A condense to a liquid.

☐

B exert zero pressure.

☐

C have maximum potential energy.

☐

D have zero electrical resistance.

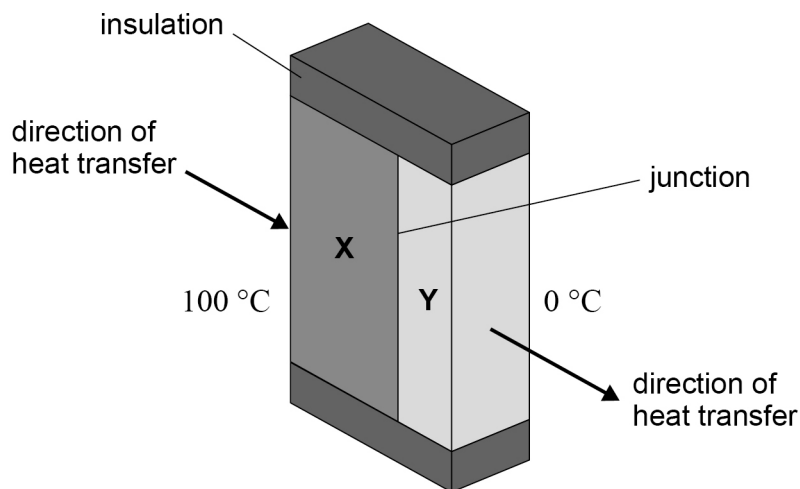
☐

0 9 Two materials **X** and **Y** of equal cross-sectional area are joined together.

The left-hand surface of **X** is at a temperature of $100\text{ }^{\circ}\text{C}$. The right-hand surface of **Y** is at a temperature of $0\text{ }^{\circ}\text{C}$.

The rates of heat transfer through **X** and **Y** are the same.

Insulation prevents heat transfers from the top, bottom and sides of each material so that heat transfer occurs only in the direction shown.



The thermal conductivity of **X** is a quarter of the thermal conductivity of **Y**.

The thickness of **X** is twice the thickness of **Y**.

What is the temperature at the junction between **X** and **Y**?

[1 mark]

A $11\text{ }^{\circ}\text{C}$ ☐

B $25\text{ }^{\circ}\text{C}$ ☐

C $75\text{ }^{\circ}\text{C}$ ☐

D $89\text{ }^{\circ}\text{C}$ ☐

1 0

A student plans an experiment to test Charles's law for a gas.

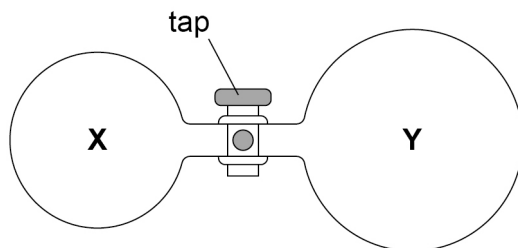
Which quantities of the gas must be measured and which must be kept constant?

[1 mark]

	Quantities to be measured	Quantities to be kept constant	
A	pressure and temperature	volume and mass	☐
B	pressure and temperature	volume	☐
C	volume and temperature	pressure and mass	☐
D	volume and temperature	pressure	☐

1 1

Bulb **X** and bulb **Y** are connected by a tube with a tap.



X has a volume of 2.0 m^3 and contains an ideal gas at an initial pressure of $4.0 \times 10^5 \text{ Pa}$ and an initial temperature of 30°C .

Y has a volume of 4.0 m^3 and initially contains no gas.

The tap is opened so that the gas in **X** and the gas in **Y** reach equilibrium.

How many gas particles are in **Y**?

[1 mark]

A 6.4×10^{25} ☐

B 9.6×10^{25} ☐

C 1.3×10^{26} ☐

D 1.9×10^{26} ☐



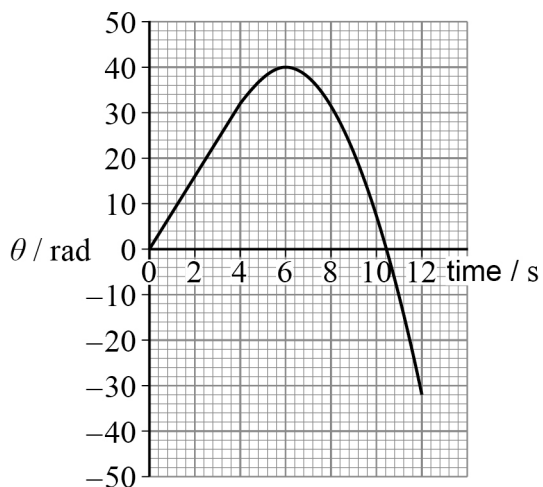
1 2 A disc rotates at a constant angular speed of 8.0 rad s^{-1} for a time of 4.0 s .

After the 4.0 s , a torque is applied to the disc. The disc has a constant angular acceleration of -4.0 rad s^{-2} .

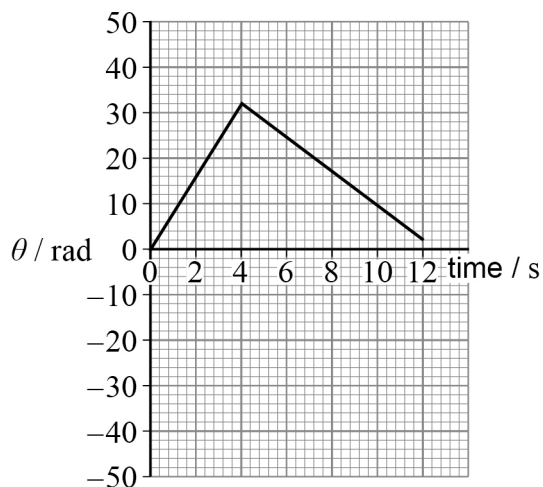
Which graph shows the variation of angular displacement θ with time?

[1 mark]

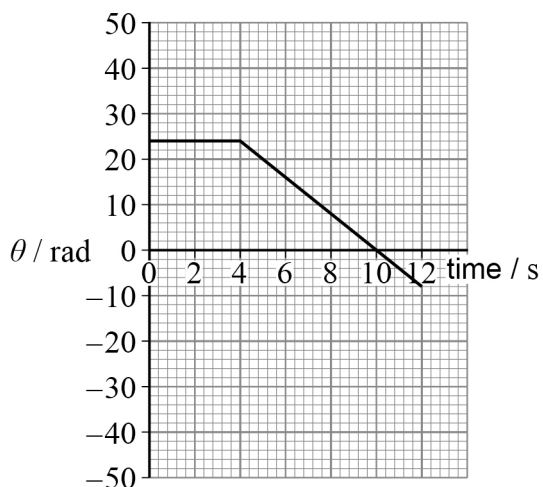
A



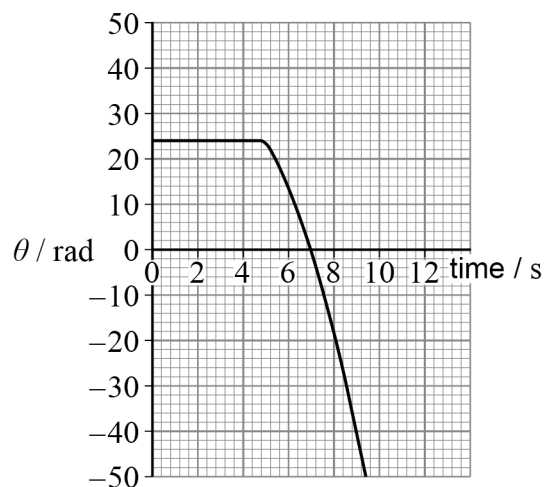
B



C



D



A ☐

B ☐

C ☐

D ☐

1 3 A graph shows the relationship between nuclear radius R and atomic number A .

Which pair of axes will produce a graph with a straight line of gradient 3?

[1 mark]

	x -axis	y -axis	
A	$\ln A$	$\ln R$	☐
B	$\ln R$	$\frac{\ln A}{3}$	☐
C	$\ln \left(\frac{A}{3} \right)$	$\ln R$	☐
D	$\ln R$	$\ln A$	☐

1 4 The purpose of the moderator in a nuclear reactor is to

[1 mark]

- A** absorb all the heat produced. ☐
- B** decrease the neutron speeds. ☐
- C** absorb excess neutrons. ☐
- D** prevent the reactor becoming critical. ☐

1 5 Which reaction is part of the hydrogen cycle in the Sun?

[1 mark]

- A** ${}_1^1\text{H} + {}_1^1\text{H} \rightarrow {}_1^2\text{H} + \text{e}^- + \bar{\nu}$ ☐
- B** ${}_1^1\text{H} + {}_1^1\text{H} \rightarrow {}_1^2\text{H} + \text{e}^+ + \nu$ ☐
- C** ${}_1^2\text{H} + {}_2^3\text{He} \rightarrow {}_2^4\text{He} + {}_1^1\text{H}$ ☐
- D** ${}_1^2\text{H} + {}_1^2\text{H} \rightarrow {}_2^4\text{He} + \gamma$ ☐



1 6 What is angular impulse in fundamental (base) units?

[1 mark]

A kg m s^{-1} ☐

B kg m s^{-2} ☐

C $\text{kg m}^2 \text{s}^{-1}$ ☐

D $\text{kg m}^2 \text{s}^{-2}$ ☐

1 7 A wheel completes 4.5 revolutions as its angular speed changes from 3.6 rad s^{-1} to 8.2 rad s^{-1} . The angular acceleration is constant.

What is the angular acceleration of the wheel?

[1 mark]

A 0.081 rad s^{-2} ☐

B 0.96 rad s^{-2} ☐

C 1.9 rad s^{-2} ☐

D 6.0 rad s^{-2} ☐

1 8 A flywheel with moment of inertia I rotates at an initial angular speed ω .

A constant torque T on the flywheel increases the angular speed to 2ω in a time t .
The angular displacement of the flywheel increases by θ .

What is the work done on the flywheel?

[1 mark]

A $\frac{3T\omega t}{2}$ ☐

B $\frac{5I\omega^2}{2}$ ☐

C $\omega T t$ ☐

D $\frac{2I\omega\theta}{t}$ ☐

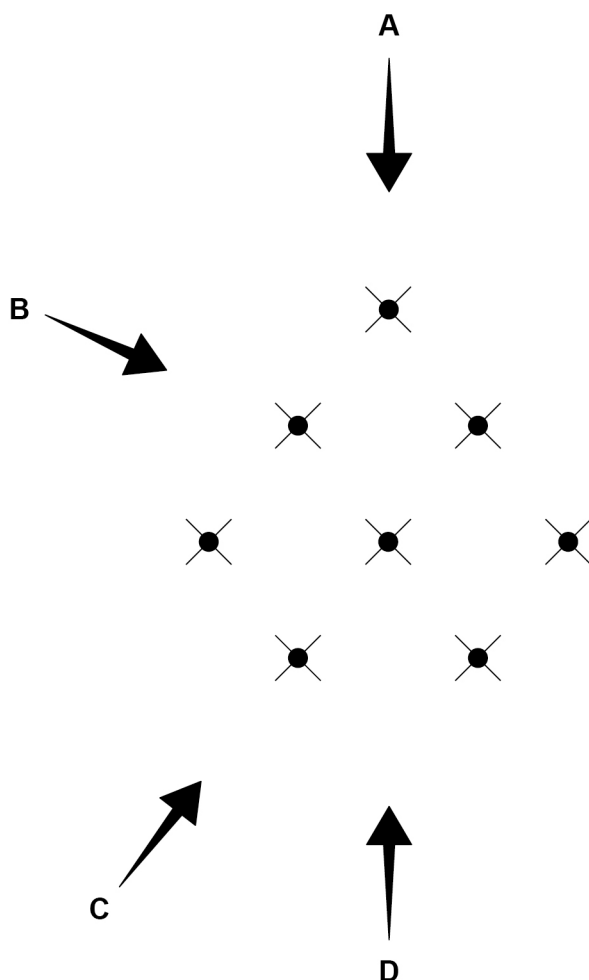


1 9 The diagram shows a plan view of an arrangement of wind turbines represented by .

The arrows represent four possible wind directions across the wind farm.

Which direction allows the maximum power to be transferred from the wind at any given wind speed?

[1 mark]



- | | |
|----------|--------------------------|
| A | ☐ |
| B | ☐ |
| C | ☐ |
| D | ☐ |



2 0

The speeds of five gas atoms are: 300 m s^{-1} , 200 m s^{-1} , 50 m s^{-1} , 150 m s^{-1} and 300 m s^{-1} .

What is c_{rms} for the atoms?

[1 mark]

A 99 m s^{-1} ☐

B 185 m s^{-1} ☐

C 200 m s^{-1} ☐

D 221 m s^{-1} ☐

2 1

The mean distance of Venus from the Sun is approximately twice the mean distance of Mercury from the Sun.

The radius of Venus is approximately 2.5 times the radius of Mercury.

What is the best estimate of $\frac{\text{total solar power incident on Venus}}{\text{total solar power incident on Mercury}}$?

[1 mark]

A $\frac{5}{8}$ ☐

B $\frac{5}{4}$ ☐

C $\frac{25}{16}$ ☐

D $\frac{25}{8}$ ☐

Turn over for the next question

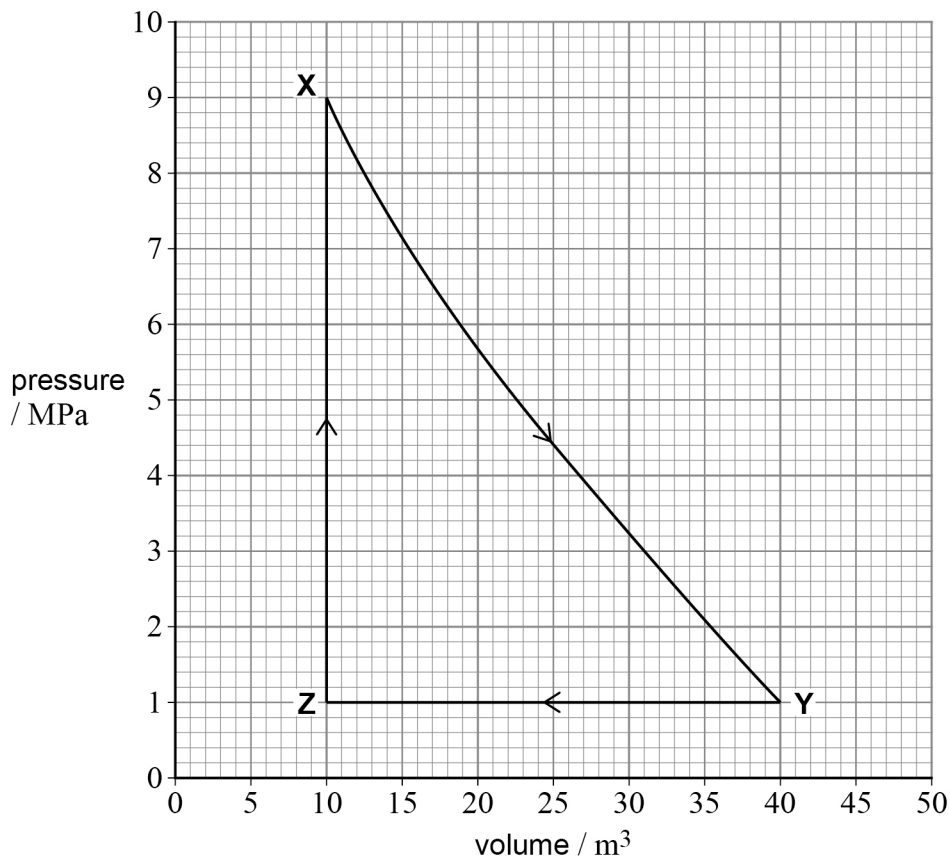
Turn over ►



2 2

Do not write
outside the
box

An ideal gas goes through a cycle of changes **XYZX** as shown in the pressure–volume graph.



What is the net work done during the cycle **XYZX**?

[1 mark]

- A** 1.2×10^8 J **on** the gas ☐
- B** 1.2×10^8 J **by** the gas ☐
- C** 1.5×10^8 J **on** the gas ☐
- D** 1.5×10^8 J **by** the gas ☐

[illegible]

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

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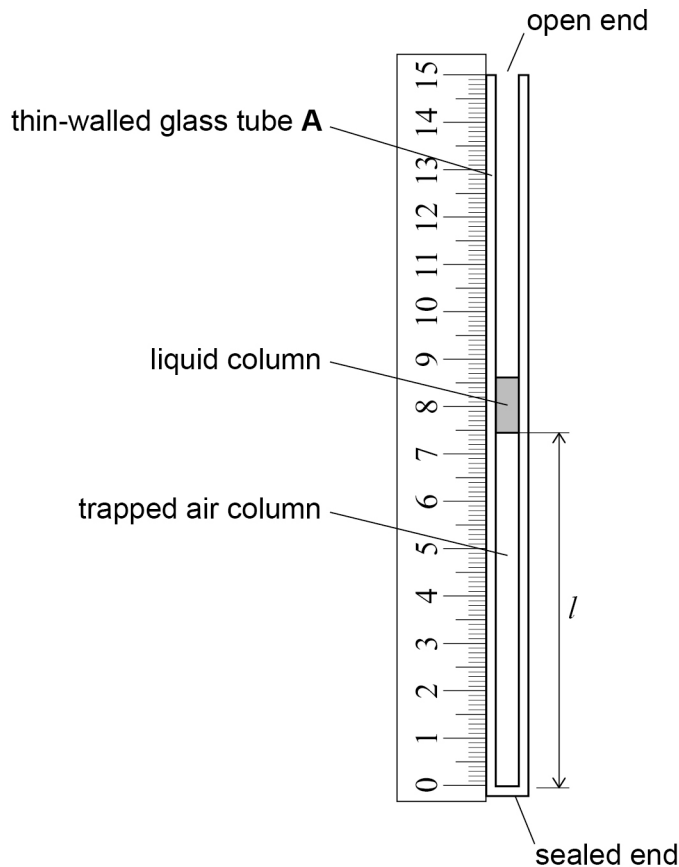


2 1 1 X P H 0 4

Figure 7 shows apparatus used to estimate absolute zero in a school laboratory.

The apparatus consists of a thin-walled glass tube **A** attached to a ruler. The tube is sealed at one end and open at the other. The tube contains a fixed mass of dry air that is trapped by a short column of liquid. The pressure of the trapped air is constant. The trapped air column has a length l .

Figure 7



The apparatus is heated slowly in a water bath. l is measured for a range of temperatures T of the water bath.

05.1

Explain why l increases as T increases.

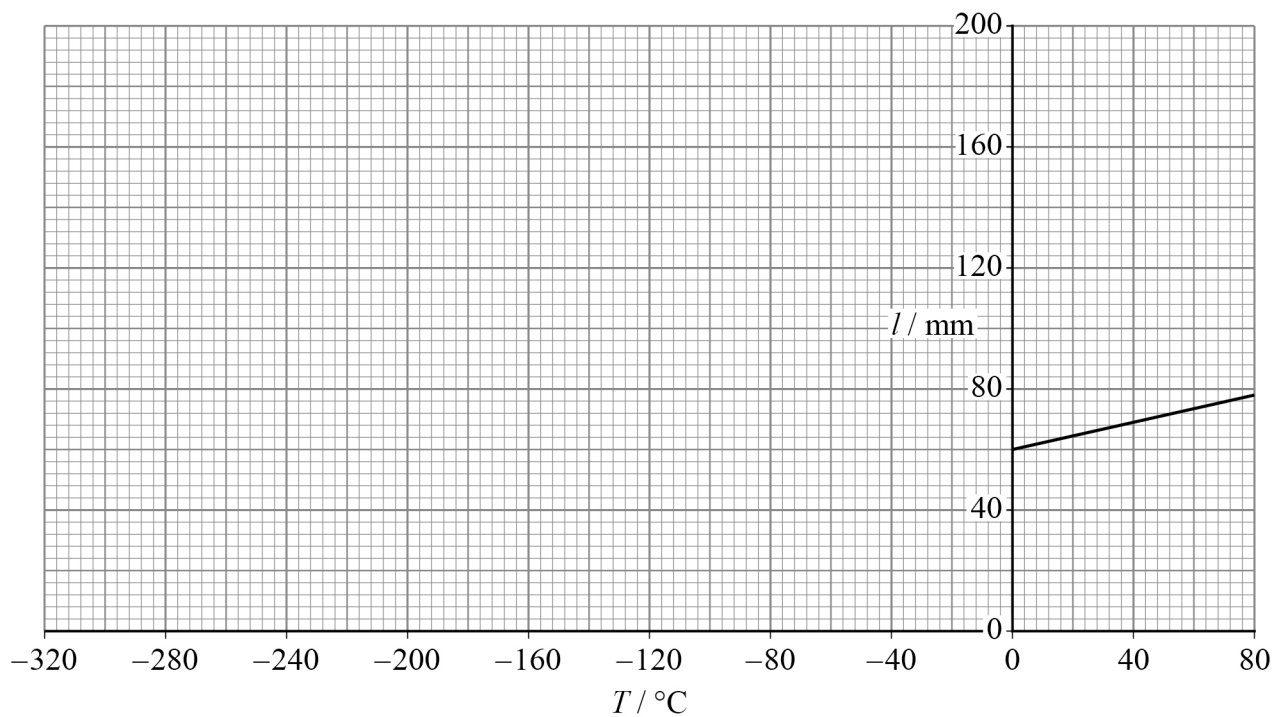
[2 marks]

Question 5 continues on the next page

Turn over ►

Figure 8 shows the variation of l with T .

Figure 8



0 5 . 2 Estimate, using **Figure 8**, absolute zero in $^\circ\text{C}$.

[1 mark]

absolute zero = _____ $^\circ\text{C}$



The trapped air contains 2.06×10^{-6} mol of dry air at a pressure of 1.01×10^5 Pa.

0 5 . 3

Calculate, in mm^2 , the internal cross-sectional area of the glass tube.
Use your estimate of absolute zero from Question **05.2** in your calculation.

[4 marks]

internal cross-sectional area = _____ mm^2

0 5 . 4

Describe **two** features of the experiment that may lead to inaccuracy when determining a value of absolute zero.

[2 marks]

1 _____

2 _____

0 5 . 5

A student suggests repeating the same experiment with a glass tube **B**.

B has half the internal cross-sectional area of **A**.

B contains the same mass of trapped air as **A**.

Draw, on **Figure 8**, the variation of l with T for **B**.

Label this line **B**.

[2 marks]



0	6	.	1
---	---	---	---

Nuclear radii can be estimated using the closest approach of an alpha particle.
Nuclear radii can be determined using electron diffraction.

Discuss the benefits of using electron diffraction rather than the closest approach of an alpha particle to evaluate nuclear radii.

[3 marks]

[illegible]

0	6	.	2
---	---	---	---

Show that the relationship between nuclear radius R and nucleon number A is given by

$$R = \left(\frac{3u}{4\pi\rho} \right)^{\frac{1}{3}} A^{\frac{1}{3}}$$

where ρ is the density of nuclear material.

[2 marks]

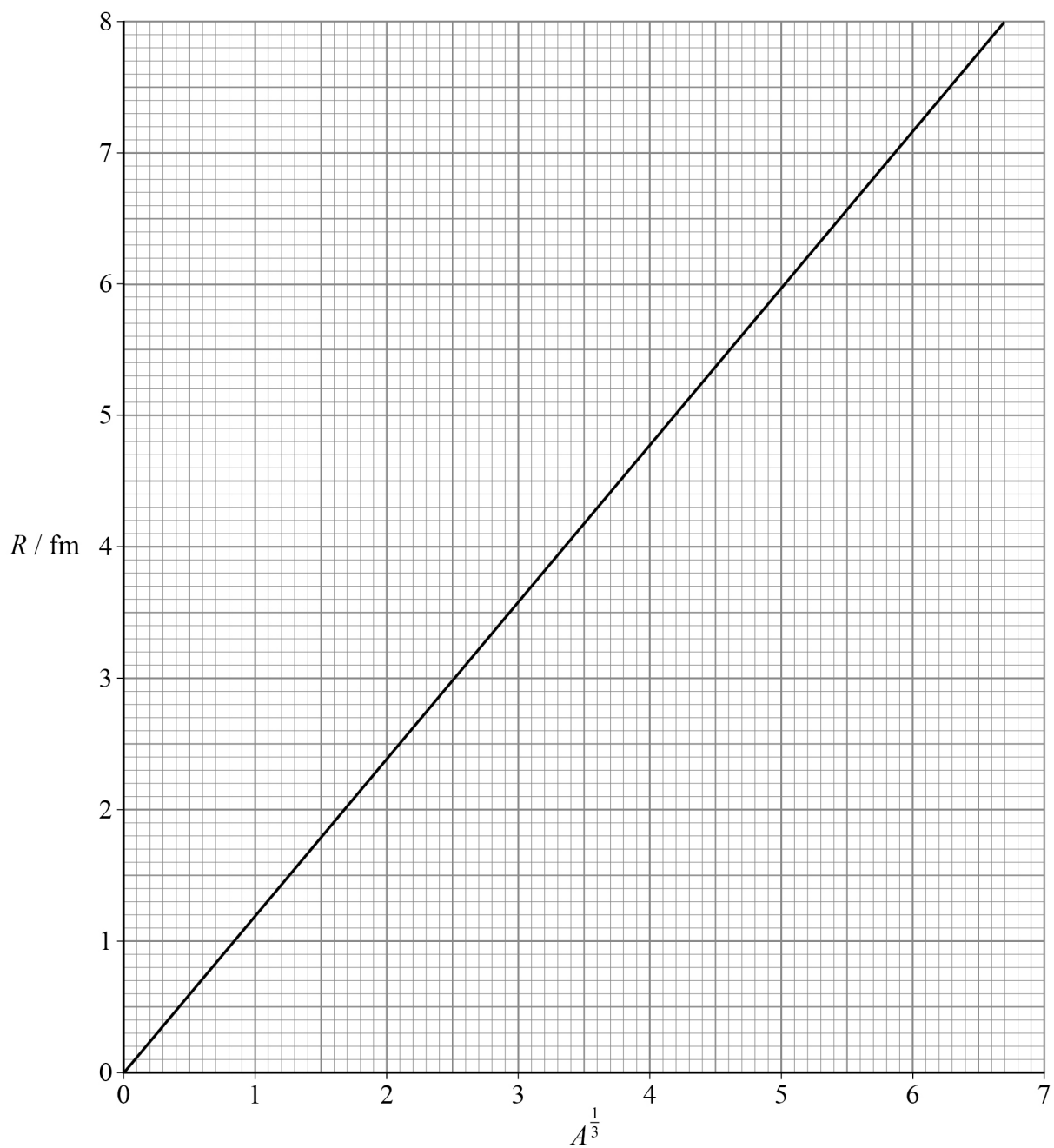
Question 6 continues on the next page

Turn over ►



Figure 9 shows the variation of R with $A^{\frac{1}{3}}$.

Figure 9



0 6 . 3 Show that ρ is approximately $2 \times 10^{17} \text{ kg m}^{-3}$.

[4 marks]

9

Turn over for the next question

Turn over ►

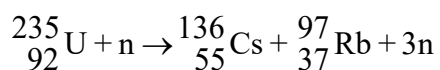


07.1

State what is meant by the binding energy of a nucleus.

[1 mark]

A possible fission of U-235 can be represented by:

**Table 1** gives the binding energy per nucleon for each nuclide in the fission.**Table 1**

Nuclide	Binding energy per nucleon / MeV
U-235	7.59094
Cs-136	8.38981
Rb-97	8.37460

07.2

Calculate, in J, the energy released in this reaction.

[3 marks]

energy released = _____ J



0 7 . 3

The average energy released by the fission of a U-235 nucleus in a reactor core is 3.2×10^{-11} J.

The fission of U-235 generates a power of 1.5 GW in the core.

Calculate the average mass of U-235 that undergoes fission in one second in this reactor core.

molar mass of U-235 = 235 g

[3 marks]

average mass = _____ kg

0 7 . 4

The thermal neutrons in the core can be modelled as the particles of an ideal gas. The effective temperature of this gas is 850 K.

Calculate c_{rms} for the thermal neutrons for this model.

[3 marks]

c_{rms} = _____ m s^{-1}

Question 7 continues on the next page

Turn over ►



07.5

Explain the role of control rods in ensuring a constant power output in a nuclear reactor core.

[3 marks]

END OF SECTION A



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

0 8 Which row shows a possible combination of changes as energy is added to a fixed mass of an ideal gas?

[1 mark]

	Work done on the gas ΔW	Gain in internal energy of the gas ΔU	Change in pressure of the gas Δp	
A	zero	positive	zero	☐
B	zero	positive	positive	☐
C	positive	zero	zero	☐
D	positive	zero	negative	☐



09

What are the fundamental (base) units for thermal conductivity?

[1 mark]

A $\text{kg m s}^{-3} \text{K}^{-1}$

☐

B $\text{kg m s}^{-2} \text{K}^{-1}$

☐

C $\text{J m}^{-1} \text{s}^{-1} \text{K}^{-1}$

☐

D $\text{J m}^{-1} \text{K}^{-1}$

☐

10

Two hollow cubes **M** and **N** are made of the same material and are in the same room. They are filled with hot liquid at the same initial temperature.

M has sides of length x and walls of thickness t .

N has sides of length $2x$ and walls of thickness $2t$.

R is the initial rate of energy transfer by conduction from **M**.

What is the initial rate of energy transfer by conduction from **N**?

[1 mark]

A $8R$

☐

B $4R$

☐

C $2R$

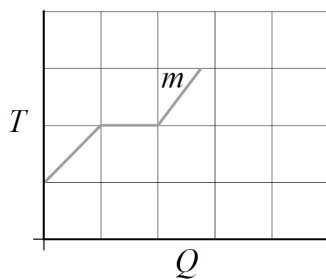
☐

D R

☐


1 1

Energy Q is transferred to a mass m of wax that is initially solid.
The graph shows the variation of the temperature T of the wax with Q .

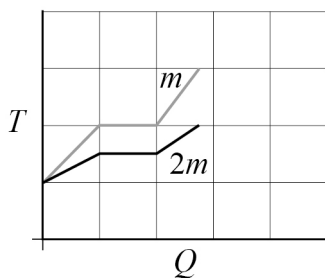


The process is repeated with wax of mass $2m$ at the same initial temperature.

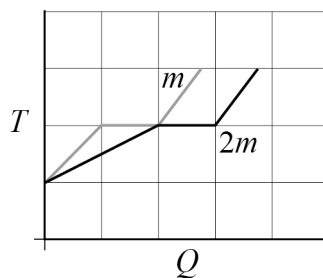
What is the variation of T with Q for this new mass?

[1 mark]

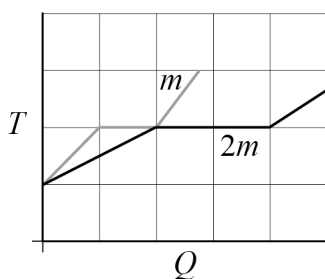
A



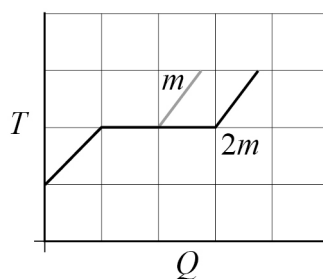
B



C



D



A

☐

B

☐

C

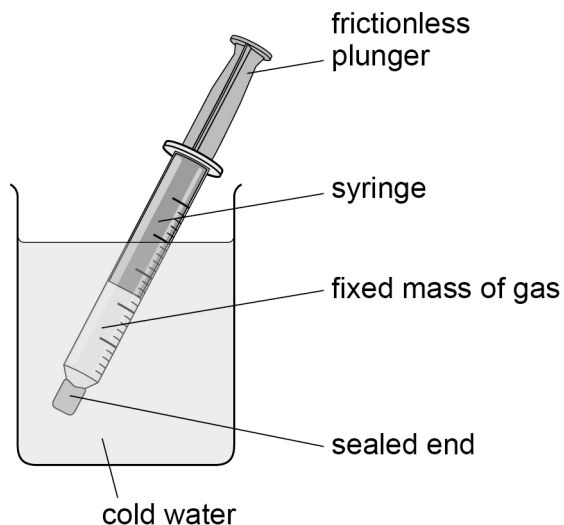
☐

D

☐

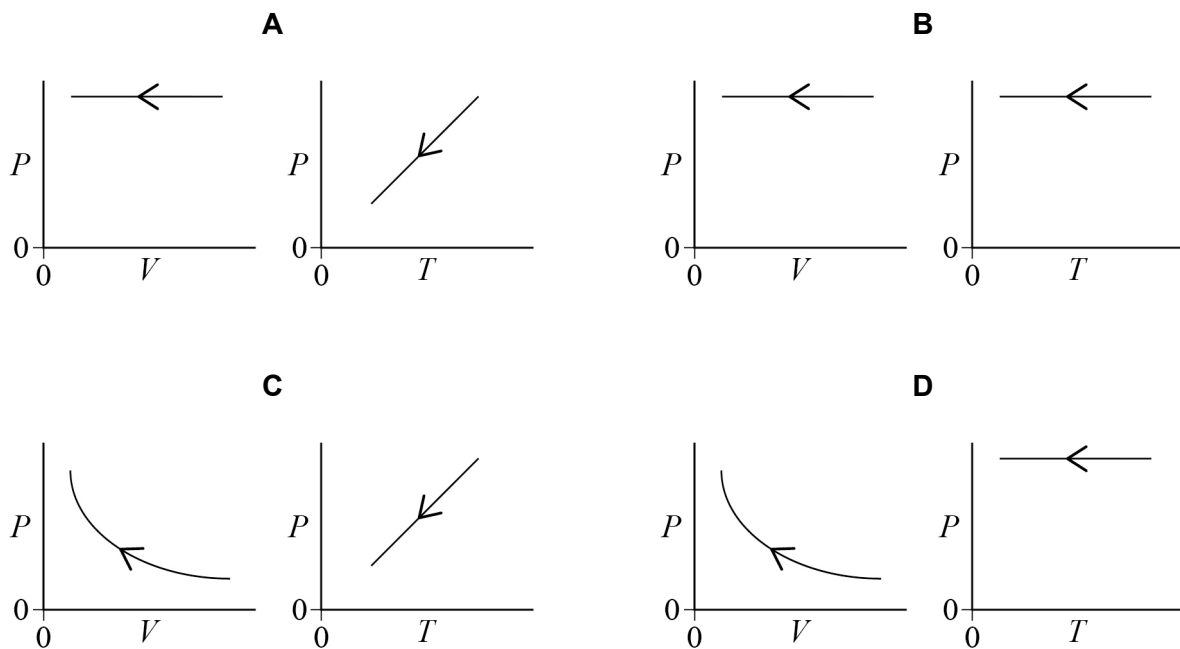
1 2 A sealed syringe with a frictionless plunger contains a fixed mass of an ideal gas.

The syringe is placed into water that is colder than the initial temperature of the gas.



Which graphs show the variations of pressure P with volume V and with absolute temperature T for the fixed mass of gas in the syringe?

[1 mark]



- A** ☐
- B** ☐
- C** ☐
- D** ☐



1 3 The speeds of three gas particles are

475 m s⁻¹ 455 m s⁻¹ 320 m s⁻¹

What is c_{rms} (root mean squared speed) for these gas particles?

[1 mark]

A 455 m s⁻¹

☐

B 422 m s⁻¹

☐

C 417 m s⁻¹

☐

D 414 m s⁻¹

☐

1 4 Which row shows desirable properties for a material chosen for the moderator in a thermal nuclear reactor?

[1 mark]

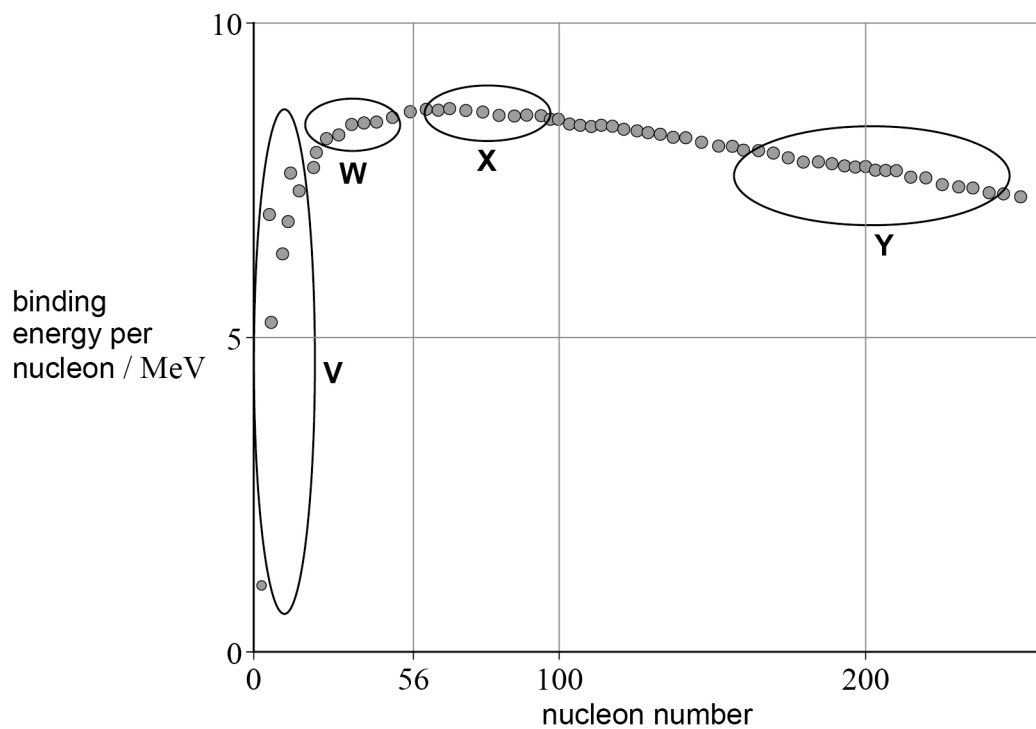
A	small nucleon number	poor neutron absorber	☐
B	small nucleon number	good neutron absorber	☐
C	large nucleon number	poor neutron absorber	☐
D	large nucleon number	good neutron absorber	☐

Turn over for the next question

Turn over ►



1 5 The plot shows the variation of binding energy per nucleon with nucleon number.



Identify the regions on the plot where nuclei will release energy when undergoing fission or fusion.

[1 mark]

	Region(s) where fusion of nuclei releases energy	Region(s) where fission of nuclei releases energy
A	W only	Y only
B	V and W	Y only
C	V only	X and Y
D	V and W	X and Y

☐
☐
☐
☐


1 6 The table shows the energy released in each reaction of the hydrogen cycle.

Reaction	Energy released / MeV
${}_1^1\text{p} + {}_1^1\text{p} \rightarrow {}_1^2\text{H} + \text{e}^+ + \nu_{\text{e}}$	0.42
${}_1^2\text{H} + {}_1^1\text{p} \rightarrow {}_2^3\text{He} + \gamma$	5.49
${}_2^3\text{He} + {}_2^3\text{He} \rightarrow {}_2^4\text{He} + 2{}_1^1\text{p}$	12.86

What is the energy released per proton when one ${}_2^4\text{He}$ nucleus is formed?

[1 mark]

A 3.12 MeV

☐

B 4.11 MeV

☐

C 4.69 MeV

☐

D 6.17 MeV

☐

1 7 Which is the most significant obstacle that must be overcome to produce a practical plasma fusion reactor?

[1 mark]

A safe disposal of the nuclear waste produced by the reactor

☐

B obtaining the deuterium fuel for the reactor

☐

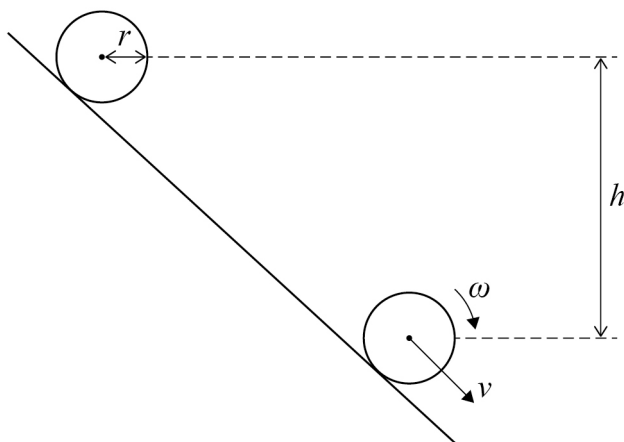
C effective confinement of the plasma in the reactor

☐

D effective emergency shut-down of the reactor

☐


- 1 8** A sphere of radius r , mass m and moment of inertia I is released from rest. It rolls down the slope without slipping.



When the sphere has moved through a vertical distance of h it has a linear speed of v and an angular speed of ω .

What is ω ?

[1 mark]

A $\sqrt{\frac{2gh}{I}}$ ☐

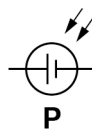
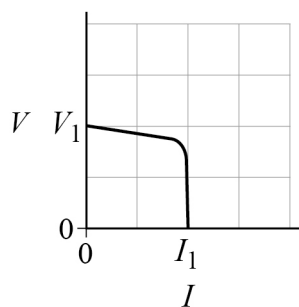
B $\sqrt{\frac{2mgh}{I}}$ ☐

C $\sqrt{\frac{m(2gh - v^2)}{I}}$ ☐

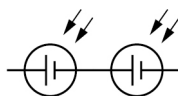
D $\sqrt{\frac{m(2gh + v^2)}{I}}$ ☐



The diagrams show the V - I characteristic and the circuit symbol for a solar cell **P**.



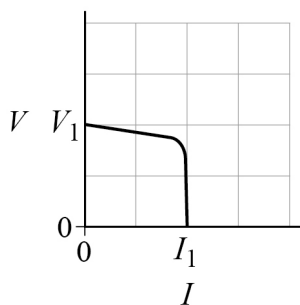
Two solar cells identical to **P** are connected in the same light conditions as **P** as shown below.



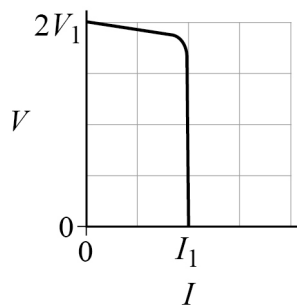
Which diagram shows the V - I characteristic for this series combination?

[1 mark]

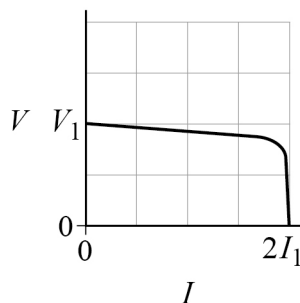
A



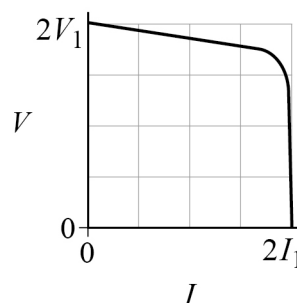
B



C



D



A



B



C



D



2 0

Which row gives an advantage and a disadvantage of the named method of electricity generation?

[1 mark]

	Method of electricity generation	Advantage	Disadvantage	
A	photovoltaic solar panels	greenhouse gases not emitted during power generation	toxic materials used in manufacture of panels	☐
B	photovoltaic solar panels	suitable for base-power stations	slow response to changes in demand	☐
C	wind turbines	not damaging to plant and animal habitats	unavailable in some weather conditions	☐
D	wind turbines	continuous production of electricity is possible	few suitable sites are available	☐



Questions 21 and 22 are about location X and location Y.

Location **X** is at sea level.

At **X**, the air has a pressure of $1.0 \times 10^5 \text{ N m}^{-2}$ and a density of 1.2 kg m^{-3} .

Location **Y** is on a mountain.

At **Y**, the air has a pressure of $0.90 \times 10^5 \text{ N m}^{-2}$ and a density of 1.1 kg m^{-3} .

Assume that the air is an ideal gas.

2 1 The air has a temperature of 300 K at **X**.

What is the temperature of the air at **Y**?

[1 mark]

A 248 K

☐

B 270 K

☐

C 275 K

☐

D 295 K

☐

2 2 The wind speed at **Y** is 50% greater than the wind speed at **X**.

P is the theoretical maximum power available to a wind turbine at **X**.

What is the theoretical maximum power available to an identical wind turbine at **Y**?

[1 mark]

A $3.7P$

☐

B $3.1P$

☐

C $1.6P$

☐

D $1.4P$

☐

[illegible]

[illegible]

Question
number

Additional page, if required.
Write the question numbers in the left-hand margin.

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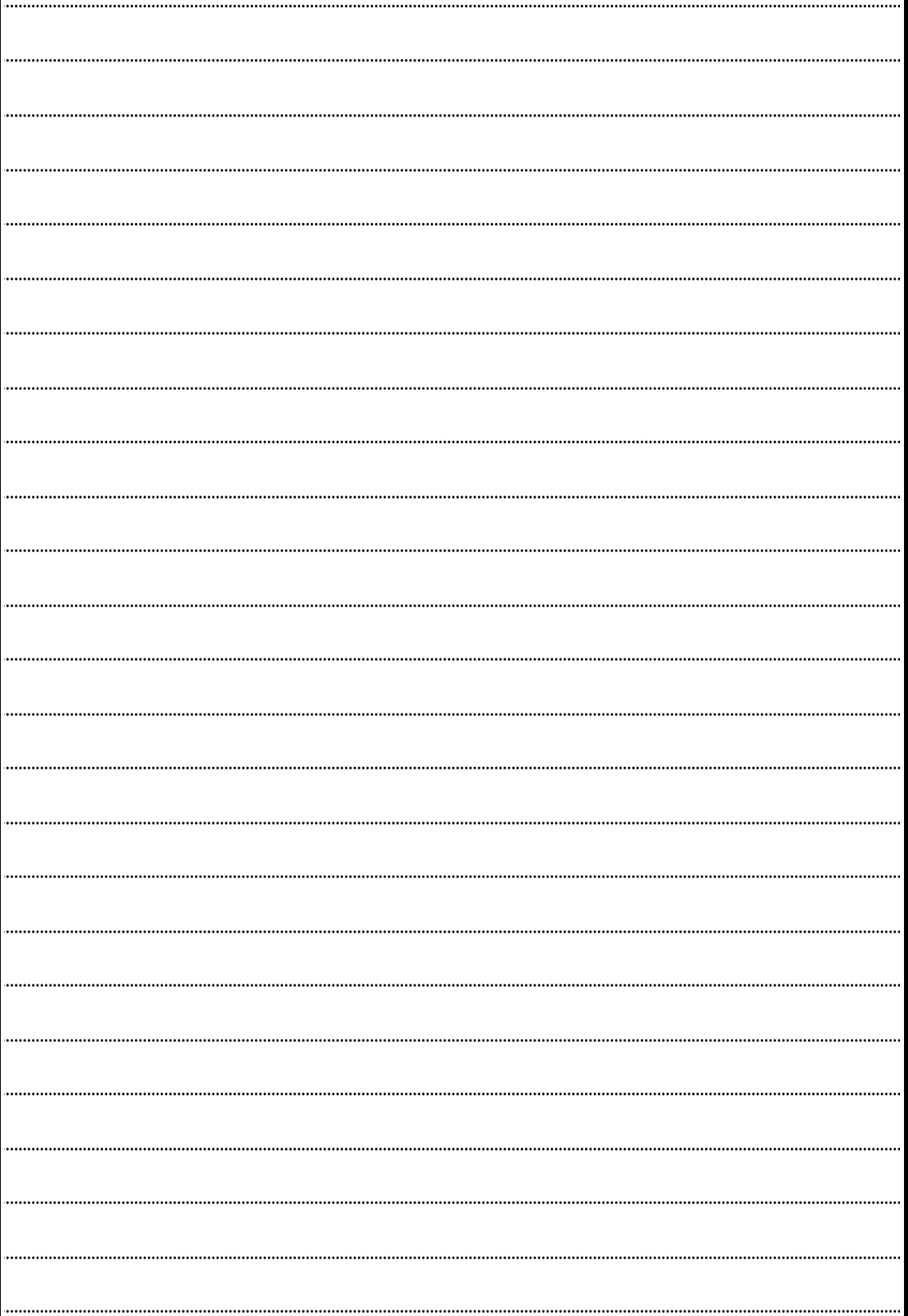
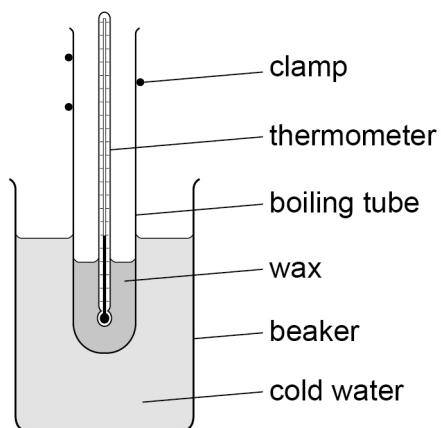


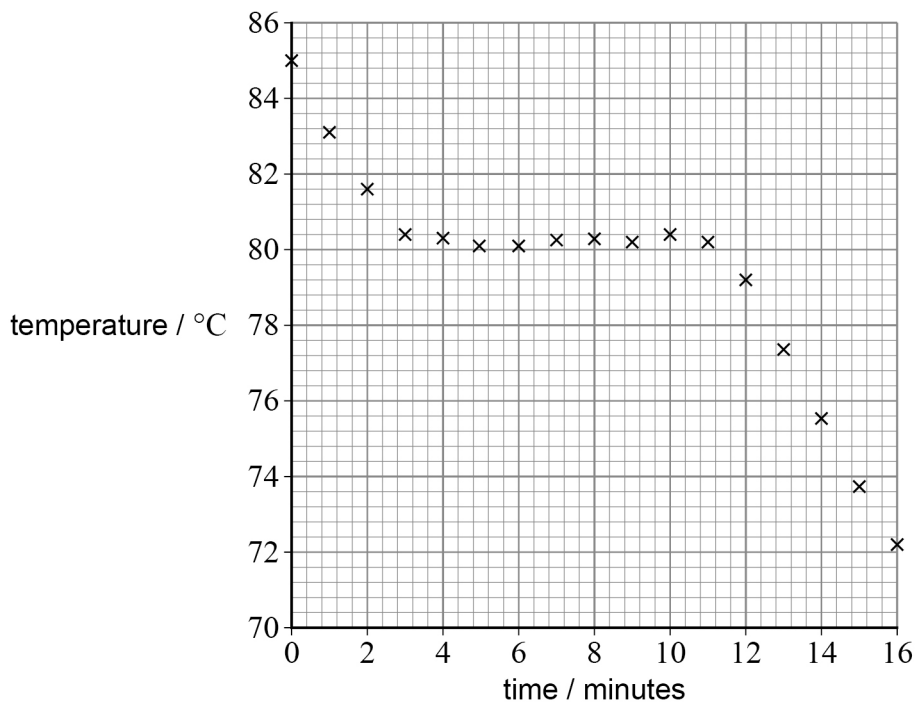
Figure 1 shows a boiling tube containing melted wax at 85°C . The boiling tube is in a beaker of cold water.

Figure 1



The temperature of the wax is recorded every minute for 16 minutes. **Figure 2** shows the results.

Figure 2



0 1 . 1 Estimate, using **Figure 2**, the melting point of the wax.

[1 mark]

melting point = _____ °C

0 1 . 2 The internal energy of the wax decreases at an average rate of 7.8 J s^{-1} during the time the wax changes state.

Estimate the mass of wax in the boiling tube.

specific latent heat of the wax = $1.5 \times 10^5 \text{ J kg}^{-1}$

[3 marks]

mass = _____ kg

0 1 . 3 Describe how the average kinetic energy and the average potential energy of the wax particles change between 8 and 16 minutes.

[2 marks]



0 2 . 1 The volume of a fixed mass of gas is kept constant.

Explain why the pressure of the gas decreases when the temperature is decreased.

[3 marks]

0 2 . 2 Show that the pressure p of an ideal gas can be given as

$$p = \frac{\rho RT}{M}$$

where ρ is the density, T is the absolute temperature and M is the molar mass of the gas.

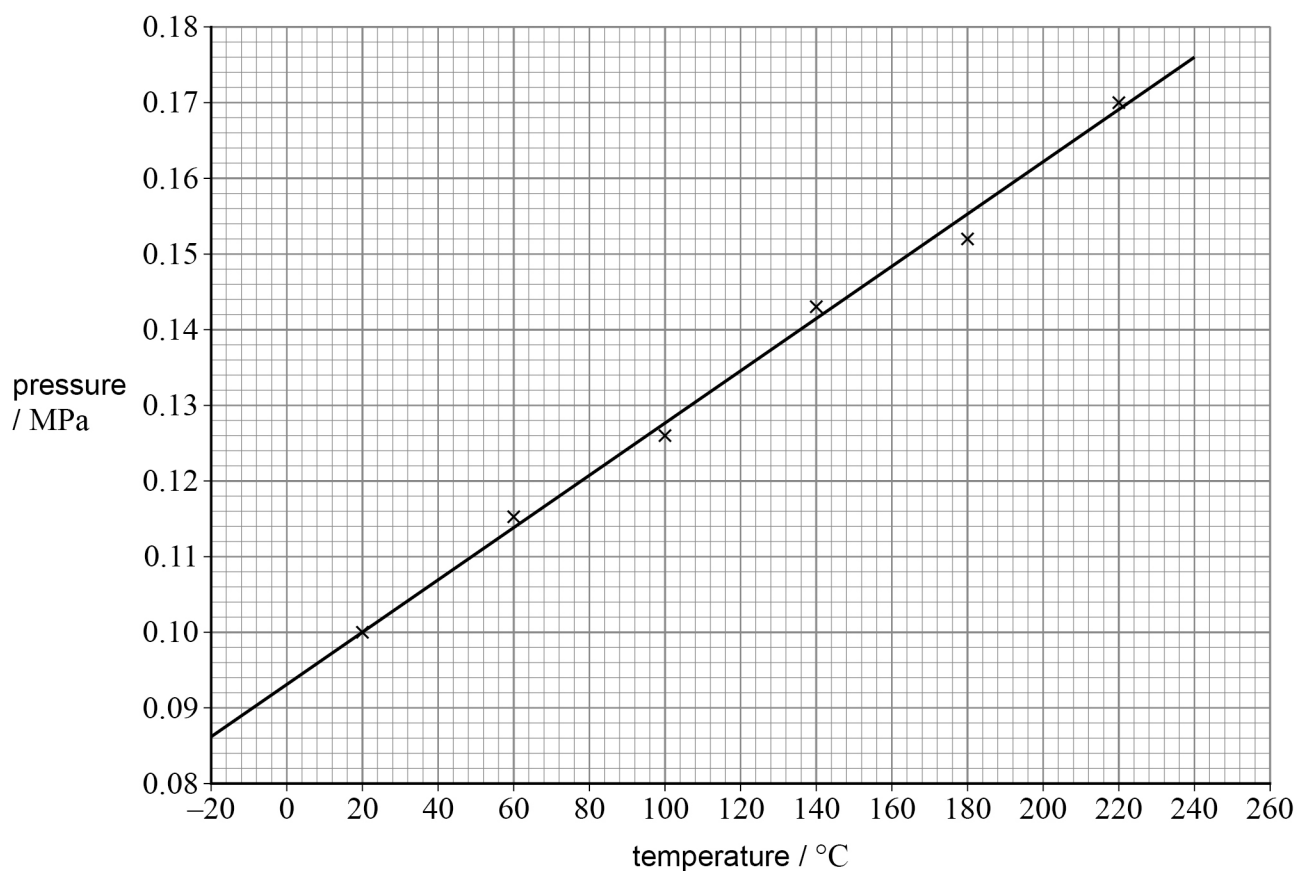
[2 marks]



0 2 . 3

Figure 3 shows the variation of pressure with temperature for a gas at a constant volume.

Figure 3



The density of the gas is 1.6 kg m^{-3} .

Assume that the gas is ideal.

Determine, in kg mol^{-1} , the molar mass of the gas.

[4 marks]

molar mass = _____ kg mol^{-1}

0 3 A flywheel energy storage system (FESS) stores kinetic energy in a rotating mass.

0 3 . 1 The flywheel in one FESS rotates at 1.4×10^4 revolutions per minute.

Show that the angular speed of the flywheel is approximately $1.5 \times 10^3 \text{ rad s}^{-1}$.

[1 mark]

0 3 . 2 The flywheel has a moment of inertia of 92 kg m^2 .

Show that the rotational kinetic energy stored by the flywheel is approximately 100 MJ.

[2 marks]

0 3 . 3 The flywheel connects to a generator. The generator exerts a constant resistive torque of 120 N m on the flywheel for 4.0 minutes.

Calculate the angular speed of the flywheel after 4.0 minutes.

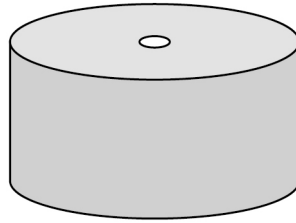
[3 marks]

angular speed = _____ rad s^{-1}



0 3 . 4 The flywheel in one FESS is a solid steel cylinder, as shown in **Figure 4**.

Figure 4



The maximum energy that can be safely stored in a FESS is limited by the material properties of the flywheel.
Designers consider replacing the steel flywheel with one of identical dimensions made of carbon fibre. Data about each material are shown in **Table 1**.

Table 1

Material	Breaking stress / GPa	Density / kg m⁻³
steel	1.5	8.0×10^3
carbon fibre	4.5	1.6×10^3

Each design is required to store the same amount of rotational kinetic energy.

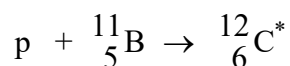
Suggest why the designers need to consider these densities and breaking stresses when choosing a material for the flywheel.
Calculations are not required.

[3 marks]

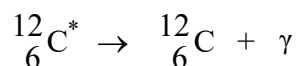


0	4
---	---

A proton with 0.675 MeV of kinetic energy collides and fuses with a stationary nucleus of boron-11 (${}^{11}_5\text{B}$). An excited carbon-12 (${}^{12}_6\text{C}^*$) nucleus is produced.



This excited carbon-12 nucleus de-excites to a stable carbon-12 (${}^{12}_6\text{C}$) nucleus by emitting a gamma photon.



0	4	.	1
---	---	---	---

Calculate, in MeV, the energy of this gamma photon.
Assume the carbon nucleus does not recoil when the gamma ray is emitted.

State your answer to an appropriate number of significant figures.

$$\text{mass of } {}^{11}_5\text{B nucleus} = 11.00666 \text{ u}$$

$$\text{mass of } {}^{12}_6\text{C nucleus} = 11.99680 \text{ u}$$

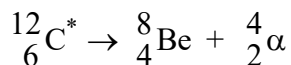
[5 marks]

energy of gamma photon = _____ MeV



0 4 . 2

An alternative mode of decay for $^{12}_6\text{C}^*$ is decay into a beryllium-8 $\left(^8_4\text{Be}\right)$ nucleus and an alpha particle $\left(^4_2\alpha\right)$.



The carbon nucleus is initially stationary. The beryllium-8 nucleus and the alpha particle move as a result of the decay.

Compare the momentum and the kinetic energy of the ^8_4Be nucleus to the momentum and the kinetic energy of the alpha particle immediately after the decay.

Support your answer with a calculation.

[4 marks]

momentum _____

kinetic energy _____

Question 4 continues on the next page

Turn over ►



0 4 . 3

One alpha particle with 2.9 MeV of kinetic energy rebounds from a different stationary $^{12}_6\text{C}$ nucleus.

Show that the closest separation between the alpha particle and the $^{12}_6\text{C}$ nucleus is approximately 6 fm.

[3 marks]

0 4 . 4

Nuclear radii were initially estimated using the technique of the closest approach of alpha particles.

Electron-diffraction experiments give more accurate determinations of nuclear radii.

Suggest why.

[2 marks]



0 5 . 1 The intensity of solar radiation at the radius of the Earth's orbit is 1400 W m^{-2} .

Calculate the power output of the Sun.

distance of Earth from Sun = $1.5 \times 10^8 \text{ km}$

[2 marks]

power output = _____ W

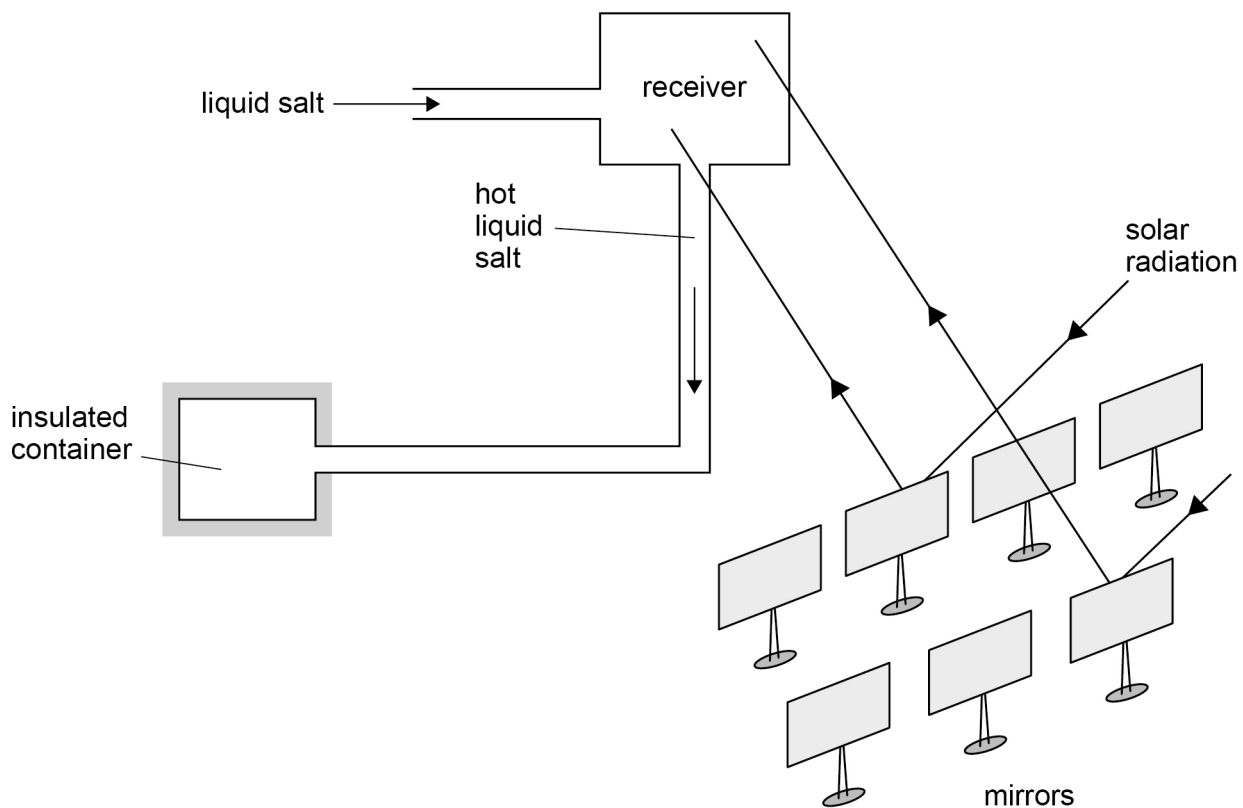
Question 5 continues on the next page



Concentrated solar power (CSP) stations store the energy from solar radiation.

Figure 5 shows one CSP station.

Figure 5



Mirrors reflect solar radiation onto a receiver. Energy from the solar radiation is transferred to liquid salt that flows through the receiver. The hot liquid salt is stored in an insulated container.



0 5 . 2 This CSP station has 10 000 mirrors. Each mirror has a reflecting area of 120 m^2 and a reflection efficiency of 90%.

The intensity of solar radiation at each mirror is 900 W m^{-2} .

Calculate, in MW, the total power of the solar radiation reflected by the mirrors.

[2 marks]

total power = _____ MW

0 5 . 3 The reflected solar radiation heats liquid salt flowing through the receiver.

Liquid salt enters the receiver at 290°C and leaves at 570°C .

The salt flows through the receiver at a rate of $0.75 \text{ m}^3 \text{ s}^{-1}$.

Calculate the rate of energy transfer to the liquid salt in the receiver.

specific heat capacity of liquid salt = $1.5 \text{ kJ kg}^{-1} \text{ K}^{-1}$

density of liquid salt = 1800 kg m^{-3}

[3 marks]

rate of energy transfer = _____ J s^{-1}

Question 5 continues on the next page

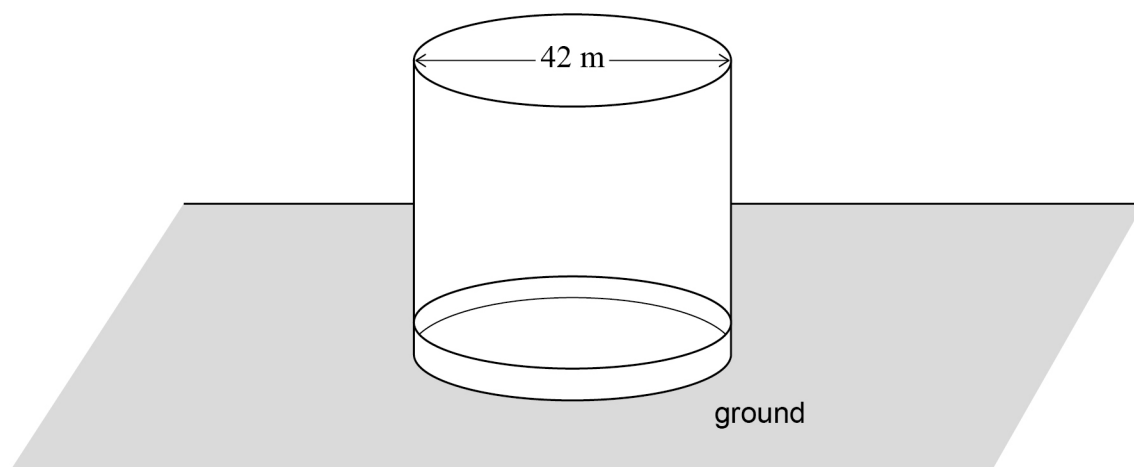
Turn over ►



0 5 . 4 The hot liquid salt is stored in an insulated container.

Figure 6 shows one cylindrical storage container for liquid salt at $570\text{ }^{\circ}\text{C}$. The internal diameter of the container is 42 m.

Figure 6



The initial rate of energy transfer through the floor of the container must be less than 100 kW. Assume that the temperature of the ground below the container remains constant at $15\text{ }^{\circ}\text{C}$.

Deduce whether a concrete floor of 45 cm thickness will be suitable.

thermal conductivity of concrete = $0.24\text{ W m}^{-1}\text{ K}^{-1}$

[3 marks]

0 6 . 1 Define the binding energy of a nucleus.

[1 mark]

0 6 . 2 Calculate the binding energy of a nucleus of thorium-232 $\left({}^{232}_{90}\text{Th}\right)$.

$$\text{mass of } {}^{232}_{90}\text{Th nucleus} = 3.852 \times 10^{-25} \text{ kg}$$

[4 marks]

binding energy = _____ J

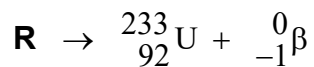
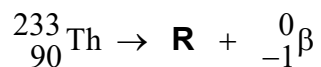
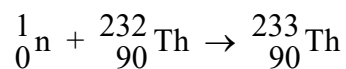
Question 6 continues on the next page

Turn over ►



06.3

Thorium-232 nuclei absorb neutrons and become uranium-233 nuclei after a series of radioactive decays.



Deduce the proton number and nucleon number of **R**.

[1 mark]

proton number = _____

nucleon number = _____



0 6 . 4 Uranium-233 nuclei undergo neutron-induced fission.

One possible fission reaction is shown below.

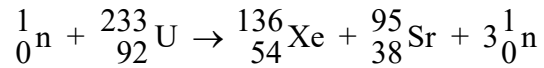


Table 2 shows the binding energy per nucleon for each nuclide.

Table 2

Nuclide	Binding energy per nucleon / MeV
${}_{92}^{233}\text{U}$	7.60396
${}_{54}^{136}\text{Xe}$	8.39619
${}_{38}^{95}\text{Sr}$	8.54912

Calculate, in J, the energy released in this fission reaction.

[3 marks]

energy released = _____ J

Question 6 continues on the next page

Turn over ►



06.5

When a large number of nuclei of uranium-233 undergo fission, the average energy released per fission will be different to your answer to Question **06.4**.

Explain why.

[1 mark]

06.6

The critical mass of uranium-233 is 16 kg.

Show that the radius of a sphere of uranium-233 of critical mass is approximately 6 cm.

$$\text{density of uranium-233} = 1.9 \times 10^4 \text{ kg m}^{-3}$$

[2 marks]



06.7 The ratio of surface area to volume for a three-dimensional object is σ , where

$$\sigma = \frac{\text{surface area}}{\text{volume}}$$

Determine σ for the sphere in Question **06.6**.

[2 marks]

$$\sigma = \quad \quad \quad \text{m}^{-1}$$

06.8 Explain why a cylinder of uranium-233 with a mass of 16 kg is unable to sustain a fission chain reaction.

[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

END OF SECTION A



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

0 7 Which produces the largest change in temperature in a system?

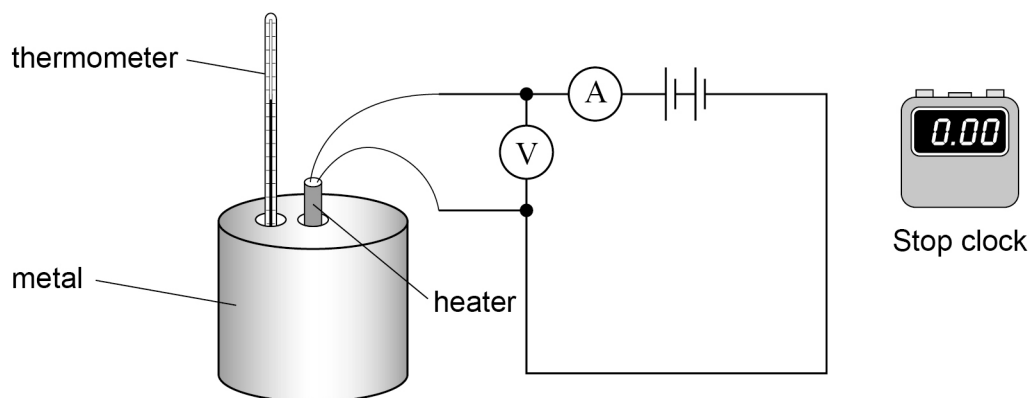
[1 mark]

	Work done on system	Heating of system	
A	+100 J	−200 J	☐
B	−100 J	+300 J	☐
C	−100 J	−300 J	☐
D	+100 J	+150 J	☐



0 8

A student uses the equipment below to determine the specific heat capacity of a metal.



Which change to the experimental procedure increases the accuracy of the measured value of the specific heat capacity?

[1 mark]

- A** insulating the metal block
- B** repeating the measurements and calculating a mean
- C** using a metal block with a greater mass
- D** using a thermometer with a smaller resolution

☐
☐
☐
☐

Turn over for the next question

Turn over ►

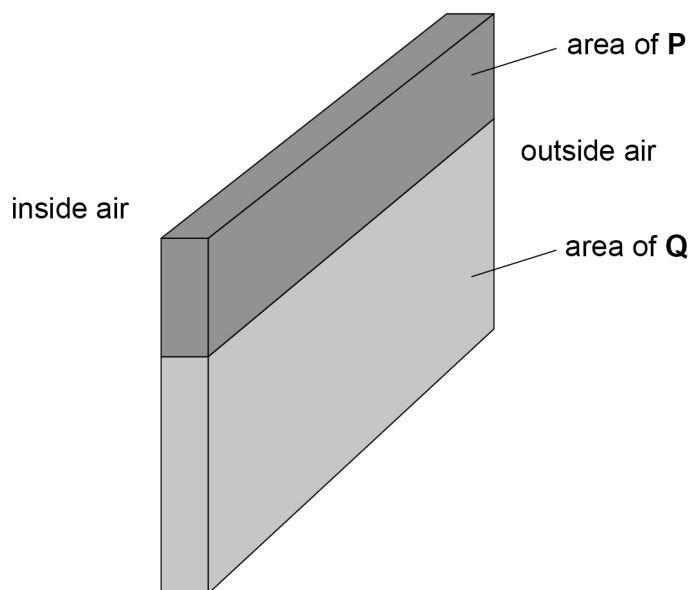


0 9

Part of a house wall contains two materials, **P** and **Q**. The wall separates air at different temperatures.

The U-value of **P** is four times the U-value of **Q**.

The area of **P** is half the area of **Q**.



What is $\frac{\text{rate of energy transfer through P}}{\text{rate of energy transfer through Q}}$?

[1 mark]

A 8

☐

B 2

☐

C $\frac{1}{2}$

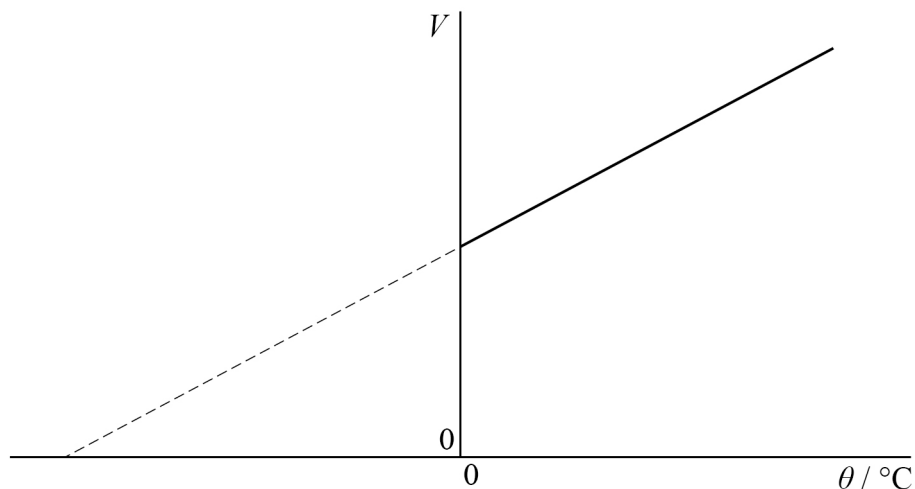
☐

D $\frac{1}{8}$

☐


1 0

The graph shows the variation of volume V with temperature θ in $^{\circ}\text{C}$ for 2 mol of an ideal gas at constant pressure p .



What is the gradient of the line?

[1 mark]

A $\frac{546R}{p}$

☐

B $\frac{2R}{p}$

☐

C $\frac{546k}{p}$

☐

D $\frac{2k}{p}$

☐

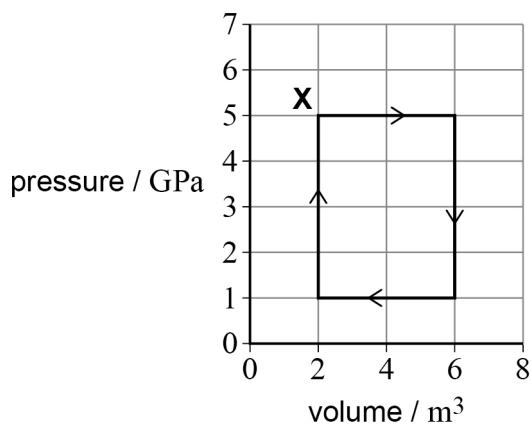
Turn over for the next question

Turn over ►



1 1

A gas undergoes one cycle of changes, starting at **X** as shown.



What is the net heat transfer to the gas?

[1 mark]

A -20 GJ

☐

B -16 GJ

☐

C +16 GJ

☐

D +20 GJ

☐

1 2

A ball of internal diameter 40 mm is filled with an ideal gas.
The pressure of the gas is 1.1×10^5 Pa and the temperature is 25 °C.

What is the number of gas particles in the ball?

[1 mark]

A 9.0×10^{20}

☐

B 7.2×10^{21}

☐

C 1.1×10^{22}

☐

D 8.5×10^{22}

☐


1 3

A sample of nitrogen gas is at $80\text{ }^{\circ}\text{C}$.
Nitrogen has a molar mass of 28 g mol^{-1} .

What is the root mean square speed of the molecules in the sample?

[1 mark]

A 0.27 km s^{-1}

☐

B 0.56 km s^{-1}

☐

C 72 km s^{-1}

☐

D 310 km s^{-1}

☐

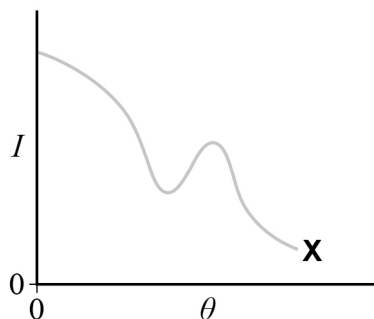
Turn over for the next question

Turn over ►



1 4

The graph shows the variation of intensity I of diffracted electrons with diffraction angle θ for a pure nuclide **X**.

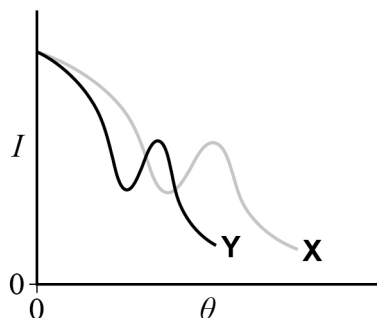


The experiment is repeated with a larger nuclide **Y**. The energy of the electrons is unchanged.

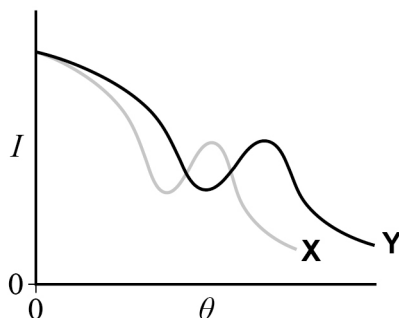
Which graph shows the variation of I with θ for nuclide **Y**?

[1 mark]

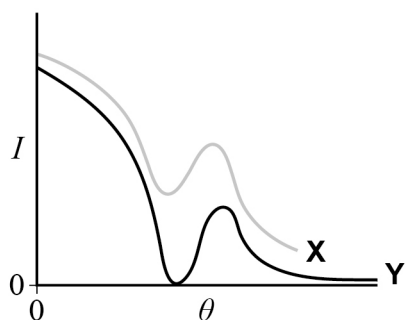
A



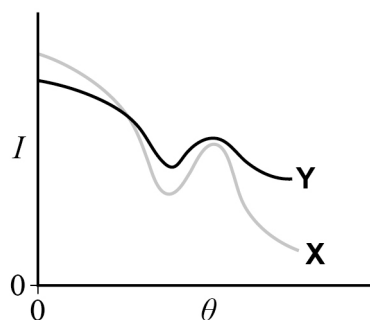
B



C



D



A



B



C



D



1 5 The radius of a polonium-210 nucleus is 5.6 fm.

What is the radius of a silicon-28 nucleus?

[1 mark]

A 0.7 fm

☐

B 1.2 fm

☐

C 2.0 fm

☐

D 2.9 fm

☐

1 6 Moderators are used in thermal nuclear reactors because

[1 mark]

A fast neutrons are less likely to escape from the reactor.

☐

B fast neutrons are more likely to decay into a proton.

☐

C slow neutrons are unaffected by control rods.

☐

D slow neutrons are more likely to be absorbed by the nuclear fuel.

☐

1 7 The first step in the hydrogen cycle for solar fusion leads to the formation of deuterium nuclei.

A deuterium nucleus is produced by the fusion of two protons and the emission of

[1 mark]

A an alpha particle.

☐

B a beta-minus particle.

☐

C a beta-plus particle.

☐

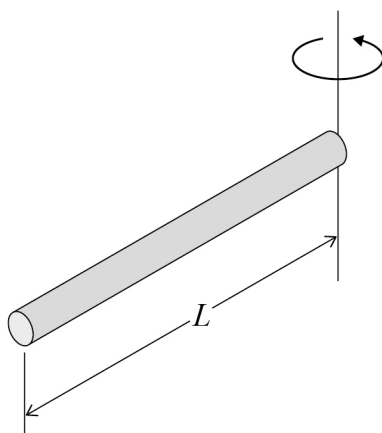
D a neutron.

☐

1 8

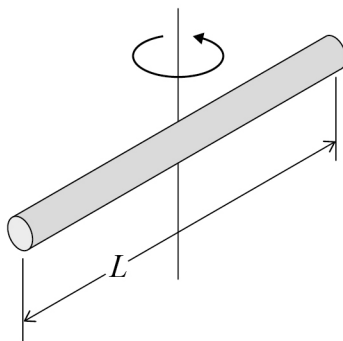
The moment of inertia of a thin rod of mass M and length L rotating about its end is $\frac{1}{3}ML^2$.

The rod rotates with angular speed ω .



The rod is now made to rotate about its centre with the same rotational kinetic energy.

The moment of inertia of the rod is now $\frac{1}{12}ML^2$.



What is the new angular speed of the rod?

[1 mark]

A $\frac{\omega}{4}$



B $\frac{\omega}{2}$



C 2ω



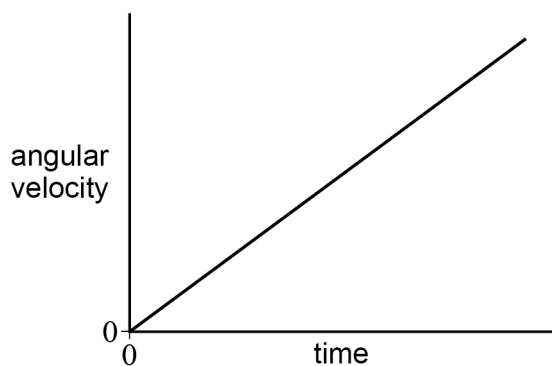
D 4ω



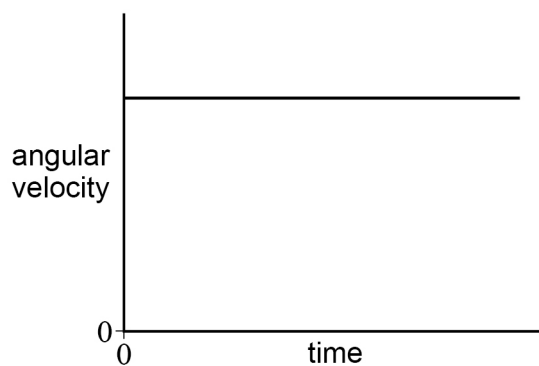
1 9 Which graph shows constant angular acceleration?

[1 mark]

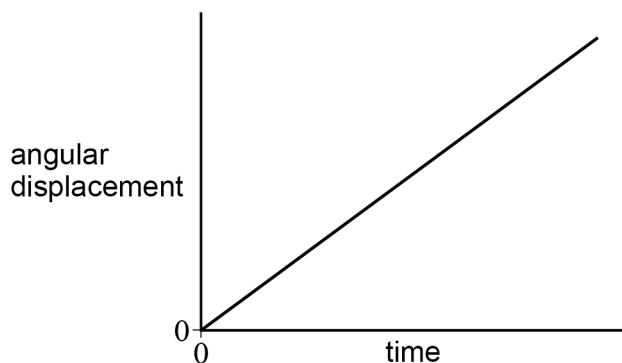
A



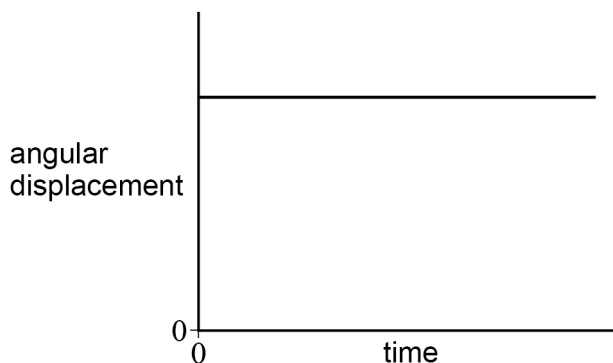
B



C



D



A

☐

B

☐

C

☐

D

☐

Turn over ►



2 0 Air moving at 12 m s^{-1} is incident on a wind turbine.

The wind turbine has a power output of 150 kW at 40% efficiency.

What length are the blades of the wind turbine?

density of air = 1.3 kg m^{-3}

[1 mark]

A 6.5 m

☐

B 8.4 m

☐

C 10 m

☐

D 21 m

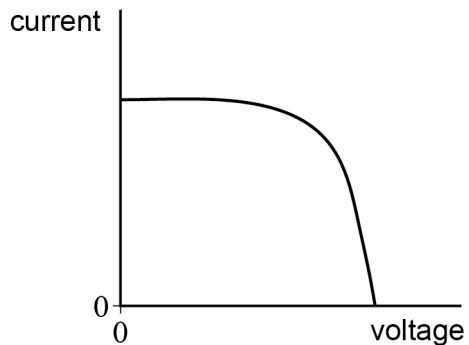
☐

2 1

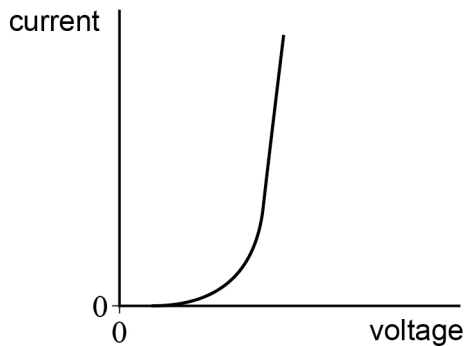
Which is the characteristic curve for an illuminated solar cell?

[1 mark]

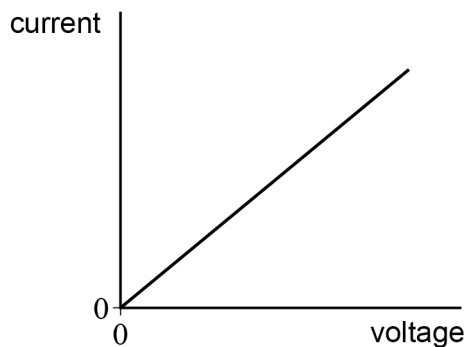
A



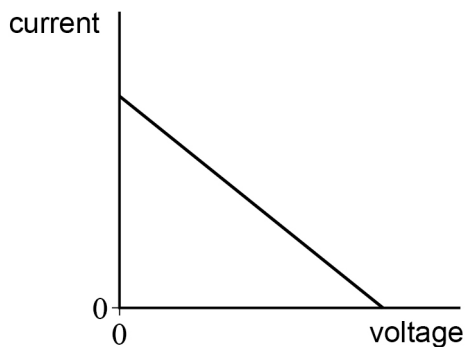
B



C



D



A

☐

B

☐

C

☐

D

☐

[illegible]

[illegible]

[illegible]

0 3 . 1

State and explain **one** practical problem in achieving fusion in reactors.

[2 marks]

0 3 . 2

The equation below describes a fusion reaction that occurs in the Sun.

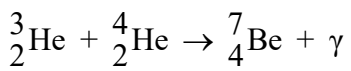


Table 2 shows the mass of each nuclide.

Table 2

Nuclide	Mass / kg
${}^3_2\text{He}$	5.00641×10^{-27}
${}^4_2\text{He}$	6.64466×10^{-27}
${}^7_4\text{Be}$	1.16482×10^{-26}

Calculate, in J, the energy released in this reaction.

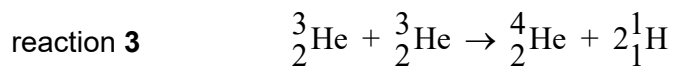
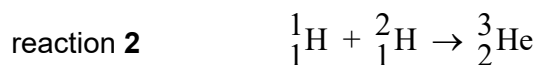
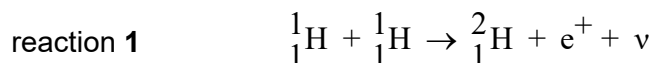
[3 marks]

energy released = _____ J

Question 3 continues on the next page



0 3 . 3 The hydrogen cycle in the Sun involves three fusion reactions:



The cycle produces one ${}_2^4\text{He}$ nucleus from four ${}_1^1\text{H}$ nuclei.

Deduce the number of times that reactions **1** and **2** occur in each cycle.

[2 marks]

Reaction **1** _____

Reaction **2** _____



Don't write

05

Uranium fuel rods used in a thermal nuclear reactor contain uranium-238 and uranium-235.

05.1

Table 3 gives the masses of a uranium-238 nuclide and a uranium-237 nuclide.

Table 3

Nuclide	Mass / u
${}^{238}_{92}\text{U}$	238.00037
${}^{237}_{92}\text{U}$	236.99831

Deduce whether a uranium-238 nucleus can decay into a uranium-237 nucleus and a neutron.

[2 marks]

05.2

Describe how a chain reaction occurs in the uranium-235 fuel.

[3 marks]



0 5 . 3 State **one** substance that is used as a moderator in a thermal nuclear reactor.

[1 mark]

0 5 . 4 The neutrons required to produce fission in many reactors are thermal neutrons.

State why they are called thermal neutrons.

[1 mark]

0 5 . 5 Explain why the kinetic energy of the neutrons needs to be reduced for the chain reaction to occur.

[2 marks]

0 5 . 6 Neutrons lose kinetic energy by a series of collisions with nuclei of the moderator. Each neutron keeps an average of 84% of its incident kinetic energy in each collision.

One neutron has an initial kinetic energy of 2.0×10^6 eV.

Deduce the average number of collisions the neutron makes to reduce its kinetic energy to 2.5×10^{-2} eV.

[2 marks]

average number of collisions = _____

Question 5 continues on the next page

Turn over ►



0 5 . 7

Liquid sodium is used as a coolant in some types of nuclear reactor.
Table 4 gives some properties of sodium.

Table 4

melting temperature / °C	98
latent heat of fusion / kJ kg ⁻¹	110
average specific heat capacity of solid / kJ kg ⁻¹ K ⁻¹	1.2
average specific heat capacity of liquid / kJ kg ⁻¹ K ⁻¹	1.3

Calculate the energy required to heat 3.3×10^5 kg of sodium from an initial temperature of 20 °C to a final temperature of 560 °C.

[3 marks]

energy = _____ J



0 5 . 8

During the operation of the reactor, radioactive caesium-137 is produced. The caesium-137 is removed from the reactor as waste which needs to be stored.

Caesium-137 decays into stable barium-137 with the emission of beta particles and gamma radiation. The half-life of caesium-137 is 30 years.

Outline the problems associated with the storage of this waste.

[2 marks]

16

Turn over for the next question

Turn over ►



Do not write Do not write

1 **6** Why is there a mass defect for a nucleus?

[1 mark]

A Smaller nuclei are always more stable than larger nuclei.

☐

B Nucleons have less potential energy in the nucleus than as separate nucleons.

☐

C A proton has a smaller mass than a neutron.

☐

D There is electrostatic repulsion between protons.

☐

Don't write

The nuclide with the largest magnitude of binding energy per nucleon is

[1 mark]

A ${}^2_1\text{H}$.



B ${}^4_2\text{He}$.



C ${}^{62}_{28}\text{Ni}$.



D ${}^{238}_{92}\text{U}$.



Don't write

0 4 . 3

The radius of the nucleus of each nuclide is estimated using the closest approach of alpha particles. The alpha particles all have the same initial kinetic energy.

Discuss how the radius estimated for **X** compares with the radius estimated for **Y**.

[3 marks]

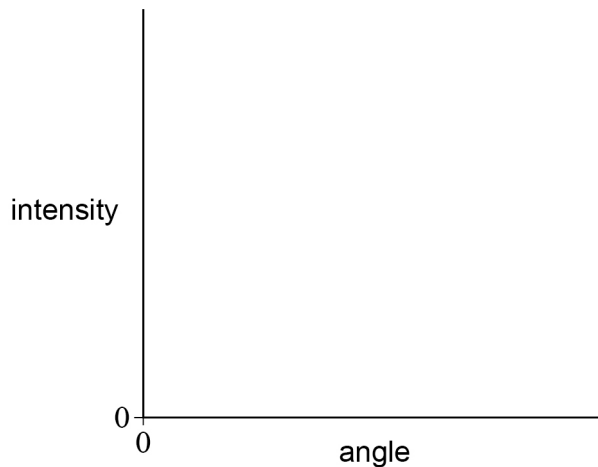
Information about the radius R of a nucleus of nucleon number A is obtained from electron diffraction.

0 4 . 4

Sketch, on the axes in **Figure 6**, the variation of intensity with angle for electron diffraction by a nucleus.

[1 mark]

Figure 6



Question 4 continues on the next page

Turn over ►



0 4 . 5

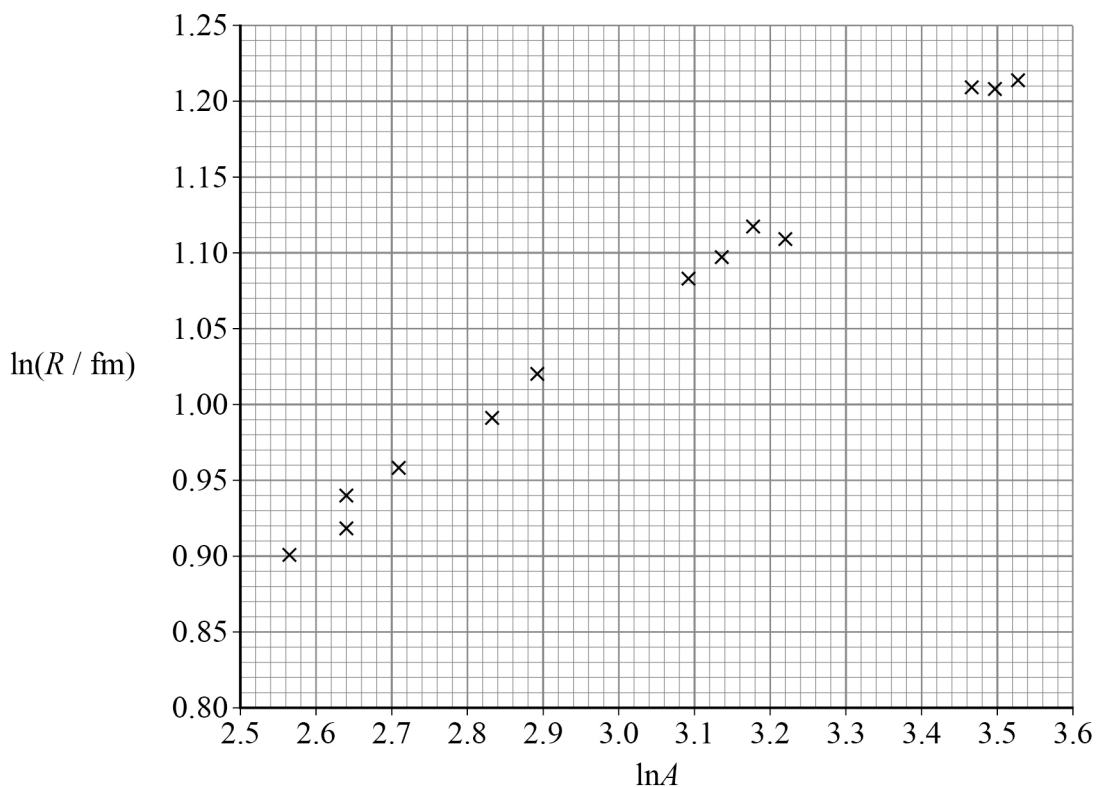
The radius of the nucleus of **X** and the radius of the nucleus of **Y** are determined using the electron-diffraction method.

Discuss how the radius determined for **X** compares with the radius determined for **Y**.

[2 marks]

Values of $\ln(R / \text{fm})$ and $\ln A$ for several nuclides are plotted on the axes in **Figure 7**.

Figure 7



0 4 . 6

The experimental data used to plot **Figure 7** suggest that R is related to A by the equation

$$R = R_0 A^{\frac{1}{3}}$$

Determine, using **Figure 7**, a value for R_0 .

[4 marks]

$R_0 =$ _____ fm

15

Turn over for the next question

Turn over ►



Do not write Do not write

07

A helium-3 (${}^3_2\text{He}$) nucleus has a binding energy of 7.72 MeV and a rest energy of 2810 MeV.

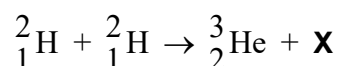
07.1

Explain why the rest energy is much greater than the binding energy.

[1 mark]

A fusion reaction between two nuclei of deuterium (${}^2_1\text{H}$) produces a helium-3 nucleus and a particle **X**.

The equation for this reaction is



3.28 MeV of energy is released in this reaction.

07.2

Determine the rest energy of **X**.

[2 marks]

rest energy = _____ MeV



0 7 . 3

Determine the binding energy of a deuterium nucleus.

[2 marks]

binding energy = _____ MeV

0 7 . 4

Many experimental fusion reactors use tritium (${}^3_1\text{H}$) and deuterium nuclei as fuel rather than two deuterium nuclei.

Discuss **two** factors that influence the choice of fuel in the design of a fusion reactor.

[2 marks]

1 _____

2 _____

7

END OF SECTION A

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

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For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



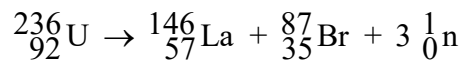
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If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.



Nuclide	Binding energy per nucleon / MeV
${}_{92}^{236}\text{U}$	7.59
${}_{57}^{146}\text{La}$	8.24
${}_{35}^{87}\text{Br}$	8.61

What is the energy released in this reaction?

[1 mark]

A 9.3 MeV

☐

B 73 MeV

☐

C 88 MeV

☐

D 161 MeV

☐

Don't write

1

3

A stable nucleus has a mass M .

The nucleus contains Z protons and A nucleons.

rest mass of proton = m_p

rest mass of neutron = m_n

What is the mass defect of the nucleus?

[1 mark]

A $Am_n + Zm_p - M$

☐

B $M - Am_n - Z(m_p - m_n)$

☐

C $Am_n + Z(m_p - m_n) - M$

☐

D $M - Am_n + Zm_p$

☐

In a thermal nuclear reactor, the number of neutrons and the speeds of neutrons need to be controlled.

What material is used to control the number of neutrons and what material is used to control the speeds of neutrons?

[1 mark]

	Control the number of neutrons	Control the speeds of neutrons	
A	graphite	water	☐
B	boron	graphite	☐
C	graphite	boron	☐
D	water	boron	☐



Do not write Do not write

A power station using a thermal nuclear reactor has an efficiency of 33%.

A fission reaction in the core results in a mass of 0.23 u being transferred into energy.

How many of these reactions per second are required to generate 750 MW of electrical power?

[1 mark]

A 1.7×10^{18}

☐

B 7.3×10^{18}

☐

C 1.5×10^{19}

☐

D 6.6×10^{19}

☐

Turn over ►



Do not write Do not write

Which statement about the critical mass of fissile material is true?

[1 mark]

A For a critical mass of fuel, each fission reaction releases one neutron.

☐

B The determination of the critical mass assumes that the fuel is spherical.

☐

C A nuclear reactor must always contain less than the critical mass of fuel.

☐

D Slowing neutrons from fission increases the amount of fuel needed to reach the critical mass.

☐

Don't write

0	4
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The solar fusion cycle has three stages.

The equation for the reaction in the third stage of the cycle is:

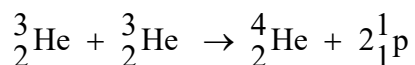


Table 1 shows the mass of each particle involved.

Table 1

Particle	Mass / u
${}^3_2\text{He}$	3.016029
${}^4_2\text{He}$	4.002603
${}^1_1\text{p}$	1.007823

0	4
---	---

1

Show that the energy equivalent of a mass of 1 u is approximately 930 MeV.

$$1 \text{ u} = 1.661 \times 10^{-27} \text{ kg}$$

[2 marks]

0	4
---	---

2

Calculate, in MeV, the energy released in one reaction in the third stage of the solar fusion cycle.

[2 marks]

energy = _____ MeV



The equation for a reaction in the second stage of the solar fusion cycle is:

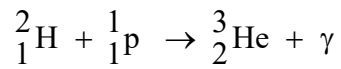


Table 2 shows the binding energy of each particle involved.

Table 2

Particle	Binding energy / pJ
${}^2_1\text{H}$	0.35593
${}^1_1\text{p}$	0
${}^3_2\text{He}$	1.23489
γ	0

0 4 . 3 Explain how the second stage contributes approximately 1.8×10^{-12} J of the energy released in one complete cycle.
Support your explanation with a calculation.

[2 marks]

0 4 . 4 Write the equation for the reaction in the first stage of the solar fusion cycle.

[2 marks]

Question 4 continues on the next page

Turn over ►



0 4 . 5

It is estimated that the Sun contains 2×10^{30} kg of hydrogen (${}^1_1\text{H}$) and emits energy at the rate of 4×10^{26} W.

Each solar fusion cycle produces 4.5×10^{-12} J of energy.

Estimate, in years, the maximum length of time for which the Sun can continue to emit energy at its current rate using only the solar fusion cycle.

mass of 1.0 mol of ${}^1_1\text{H} = 1.0 \times 10^{-3}$ kg

[4 marks]

time = _____ years

0 4 . 6

The fusion of two helium nuclei in the core of a star leads to the formation of heavier elements.

Explain why helium fusion requires the plasma temperature to be greater than the temperature needed for hydrogen fusion.

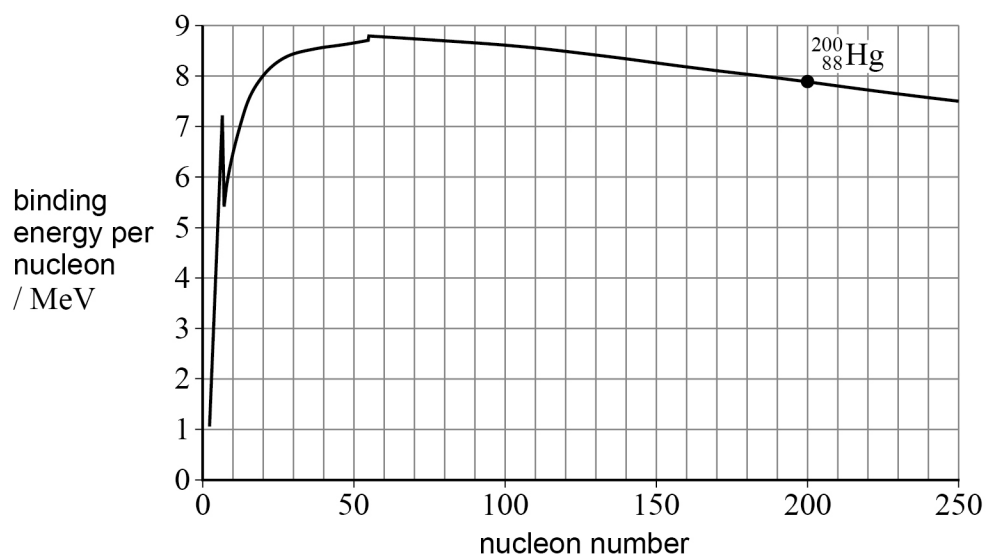
[2 marks]



Do not write

1 4 The graph shows the variation of binding energy per nucleon with nucleon number.

The position of $^{200}_{88}\text{Hg}$ is shown.



What is the best estimate of the release of energy from the fission of a $^{200}_{88}\text{Hg}$ nucleus?

[1 mark]

A 0.7 MeV ☐

B 8 MeV ☐

C 62 MeV ☐

D 140 MeV ☐



Do not write

Neutron absorption cross-section σ is measured in cm^2 . Nuclei with a large value of σ are more likely to absorb neutrons.

The table gives nucleon numbers and approximate values of σ for four isotopes.

Which isotope would be most suitable as a moderator in a thermal fission reactor?

[1 mark]

	Nucleon number	σ / cm^2	
A	6	1000	☐
B	7	0.0005	☐
C	181	0.0005	☐
D	177	1000	☐

Used fuel rods from a nuclear reactor are stored in water for a long time. The water contains dissolved boron.

Possible reasons for this type of storage are:

1. to prevent overheating due to radioactive decay of nuclei in the used fuel
2. to shield against radiation from radioactive materials in the used fuel
3. to prevent the possibility of a chain reaction occurring in the remaining uranium in the used fuel.

Which of these reasons are correct?

[1 mark]

A 1 and 2 only

☐

B 1 and 3 only

☐

C 2 and 3 only

☐

D 1 and 2 and 3

☐

Do not write

0	1
---	---

This question compares electron diffraction with alpha-particle scattering.

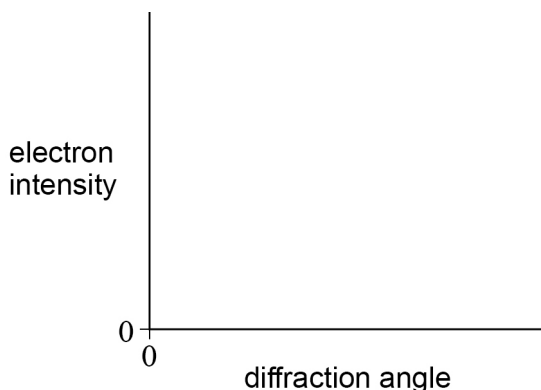
A narrow beam of electrons is passed through a thin foil of lead atoms.
The diffraction of the electrons is used to measure the diameter of lead nuclei.

0	1	.	1
---	---	---	---

Sketch, on **Figure 1**, the variation of electron intensity with diffraction angle of the electrons.

[1 mark]

Figure 1



The diameter of a lead-208 ($^{208}_{82}\text{Pb}$) nucleus is approximately equal to 11 fm.

0	1	.	2
---	---	---	---

Each electron in the beam has a momentum of $2.2 \times 10^{-19} \text{ kg m s}^{-1}$.

Deduce whether the de Broglie wavelength of these electrons is suitable for the measurement of the diameter of lead-208 nuclei.

Support your answer with a calculation.

[3 marks]



Lead-204 ($^{204}_{82}\text{Pb}$) is another isotope of lead.

0 1 . 3

Electron diffraction is now used to measure the diameter of a lead-204 nucleus.

Deduce whether using electron diffraction to measure nuclear diameter will produce different answers for $^{204}_{82}\text{Pb}$ and $^{208}_{82}\text{Pb}$ nuclei.

[1 mark]

0 1 . 4

An alpha particle with an initial kinetic energy of 8.4 MeV is aimed directly at a nucleus of $^{208}_{82}\text{Pb}$.

Assume that the $^{208}_{82}\text{Pb}$ nucleus remains stationary.

Calculate the distance of closest approach between the alpha particle and the nucleus.

[3 marks]

distance of closest approach = _____ m

0 1 . 5

The distance of closest approach is now determined with nuclei of $^{204}_{82}\text{Pb}$ using alpha particles with an initial kinetic energy of 8.4 MeV.

Deduce whether the distance of closest approach of the alpha particles will be different for $^{204}_{82}\text{Pb}$ and $^{208}_{82}\text{Pb}$ nuclei.

[1 mark]



[4 marks]

[illegible]

0	2	.	2
---	---	---	---

[3 marks]

$$\frac{c_{\text{Ar}}}{c_{\text{Xe}}} =$$



0 2 . 3

The argon atoms and the xenon atoms all contribute to the total pressure exerted by the gas in the container.

A student suggests that the xenon atoms make a greater contribution than the argon atoms to the total pressure.

Deduce whether the student is correct.

[2 marks]

9

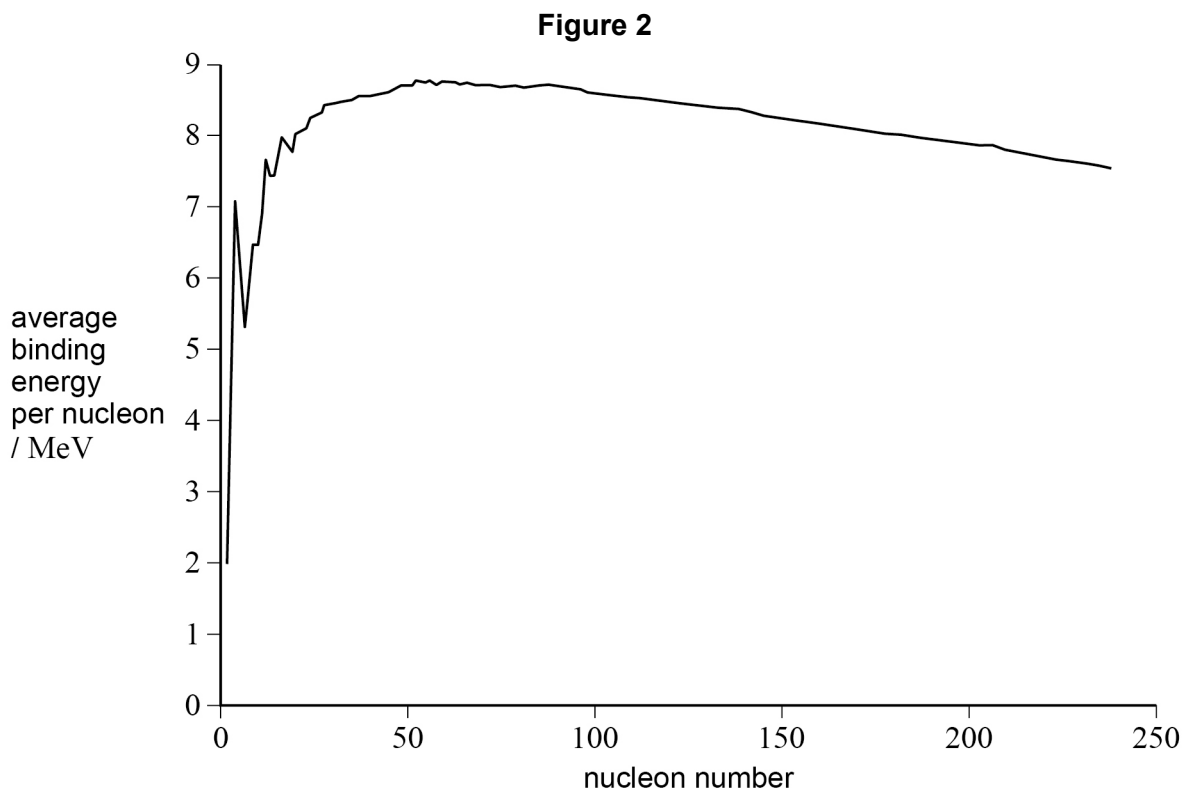
Turn over for the next question

Turn over ►



0 3 Nuclear fission and nuclear fusion can each lead to a release of energy.

0 3 . 1 **Figure 2** is a plot of the variation of average binding energy per nucleon with nucleon number.



Compare the processes of nuclear fission and nuclear fusion in which energy is released.

In your answer:

- describe the nuclear changes involved in fission and fusion
- describe how each process can begin
- explain why fission reactions are confined to one part of **Figure 2** and fusion reactions are confined to a different part of **Figure 2**.

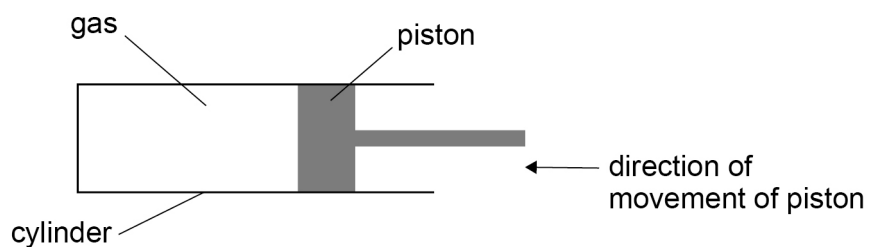
[6 marks]



[3 marks]

A cylinder with a moveable piston contains a fixed mass of an ideal gas, as shown in **Figure 3**.

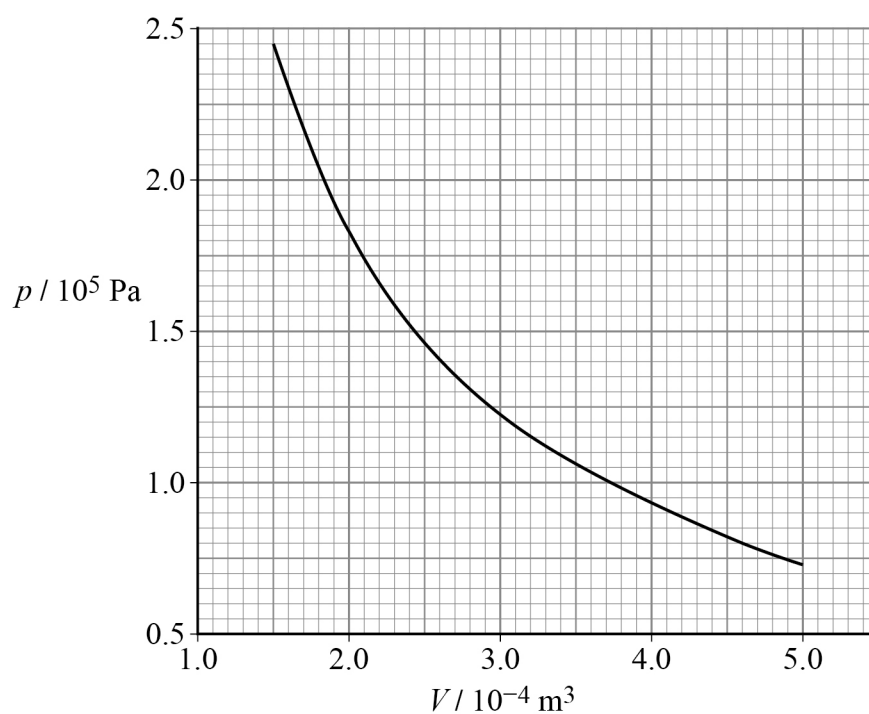
Figure 3



The gas is compressed by moving the piston to the left.

Figure 4 shows the variation of pressure p with volume V for the gas.

Figure 4



0 4 . 1

Show that, for the compression in **Figure 4**, the temperature at the start of the compression is approximately the same as the temperature at the end of the compression.

[3 marks]

0 4 . 2

Determine the work done on the gas in the compression shown in **Figure 4**.

[3 marks]

work done = _____ J

0 4 . 3

A gas is initially at the same temperature as its surroundings. The temperature of the surroundings remains constant.

Explain why it is **not** possible to compress the gas so that its temperature remains **exactly** constant, while doing work on it.

Refer to the first law of thermodynamics in your answer.

[2 marks]

Question 4 continues on the next page

Turn over ►



0 4 . 4

It is possible to compress the gas in the cylinder at a temperature that is **approximately** constant.

Describe how the compression must be done.

In your answer you should consider:

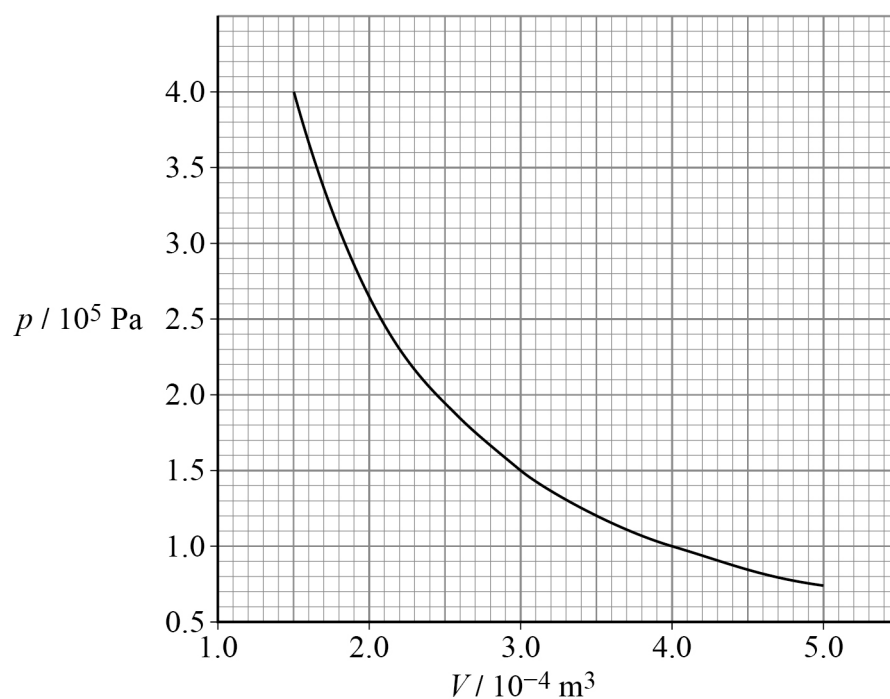
- the way in which the piston is moved
- a feature of the cylinder.

[2 marks]

The gas in **Figure 3** is now compressed in a different way so that its temperature changes significantly.

Figure 5 shows the variation of pressure p with volume V for the gas.

Figure 5



The temperature of the gas immediately after the compression shown in **Figure 5** is 223°C .

0 4 . 5

Calculate, in mol, the amount of gas in the cylinder.

[2 marks]

amount of gas = _____ mol

0 4 . 6

Calculate the temperature change of the gas during the compression shown in **Figure 5**.

[3 marks]

temperature change = _____ K

Question 4 continues on the next page



0 4 . 7

The values of p and V used to plot **Figure 5** are connected by the relationship

$$pV^b = C$$

where b and C are constants.

The values of p and V can also be processed to produce a linear graph from which b can be determined.

Suggest axes for this linear graph.

Go on to explain how the value of b can be determined from this graph.

[2 marks]



Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



Figure 6 shows an exercise bike and its drive system.

A rider turns the pedals to apply a torque to disc **A**. This torque is transmitted to hub **B** by a belt that does not slip. **B** is fixed to a larger disc **C** that rotates when the belt moves.

A spring pushes a friction pad against the edge of **C**, as shown in **Figure 7**. This causes a force F_s to act towards the centre of **C**.

Figure 6

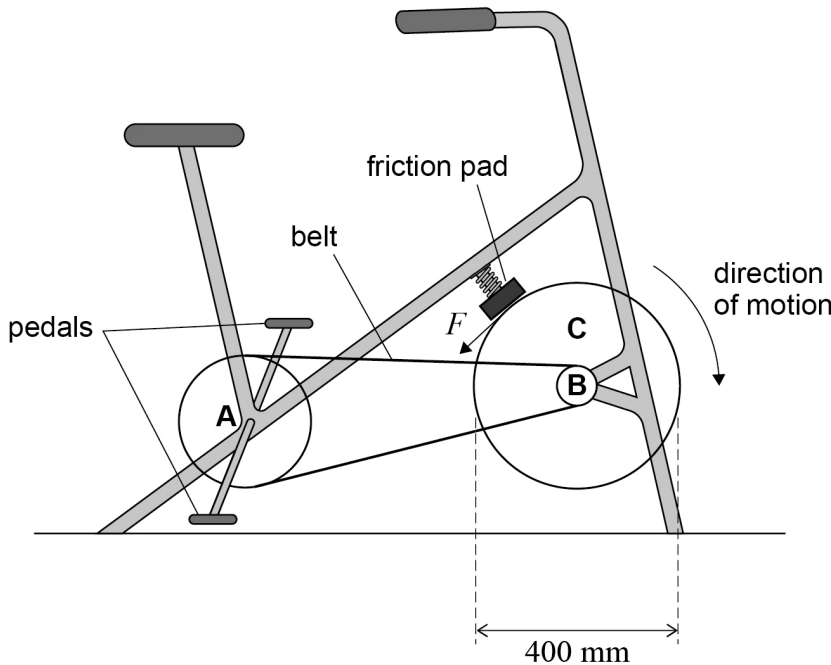
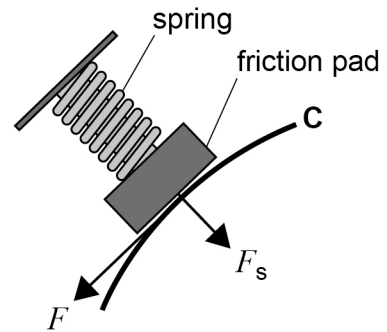


Figure 7



When **C** is turning, F_s leads to a frictional force F that acts on the surface of **C**. F produces a constant resistive torque of 4.5 N m about the centre of **C**. Assume that the drive system has an efficiency of 100%.



0 5 . 1 The magnitude of F is given by:

$$F = 0.62F_s$$

where F_s is the force in the compressed spring.

The spring constant of the spring is 2400 N m^{-1} .

C has a diameter of 400 mm.

Calculate, in m, the change in length of the spring.

[3 marks]

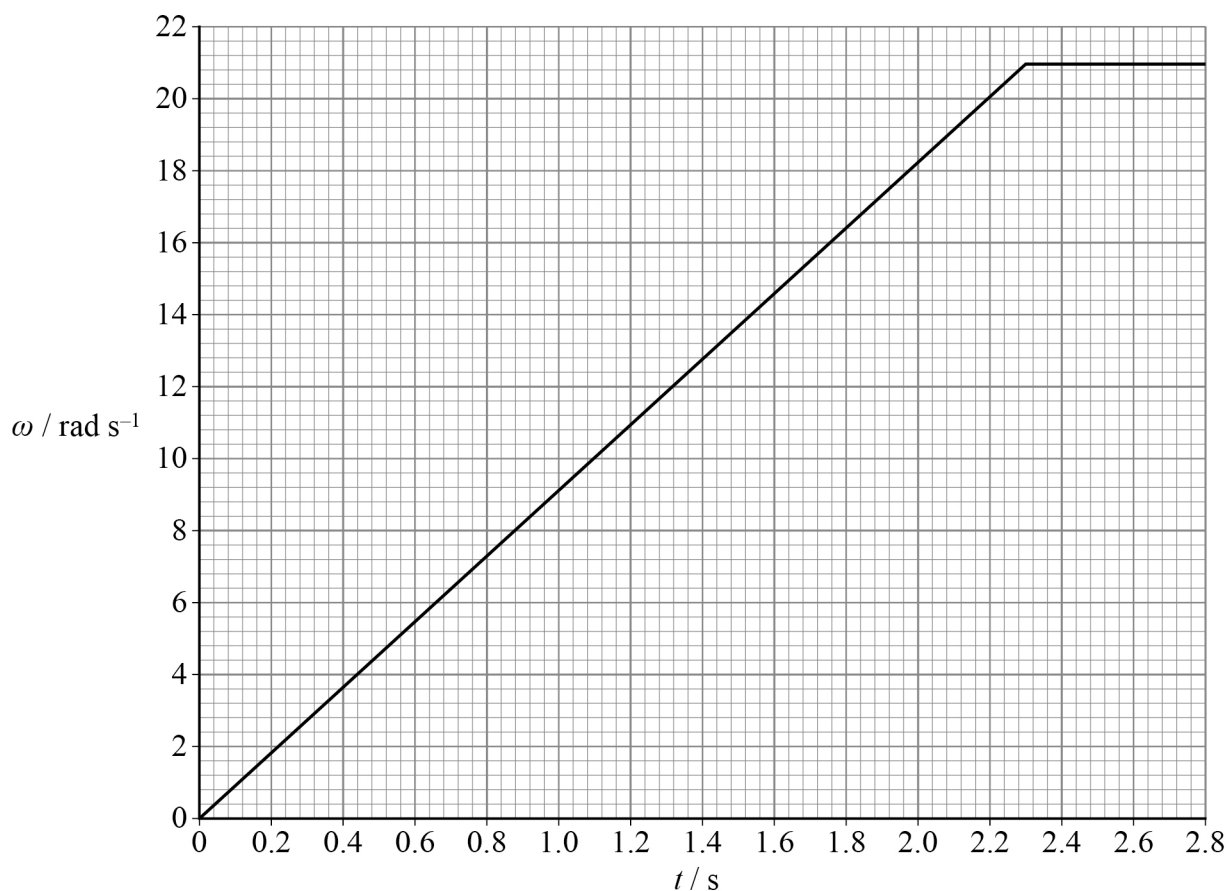
change in length = _____ m

Question 5 continues on the next page



Figure 8 shows the variation of the angular speed ω of C with time t .

Figure 8



The resistive torque due to the friction pad remains constant at 4.5 N m.

0 5 . 2

Calculate the power delivered by the rider when $t = 2.4$ s.

[1 mark]

power = _____ W



C has a moment of inertia of 0.21 kg m^2 .

0 5 . 3

Calculate the rotational kinetic energy of **C** when $t = 2.4 \text{ s}$.

[1 mark]

rotational kinetic energy = _____ J

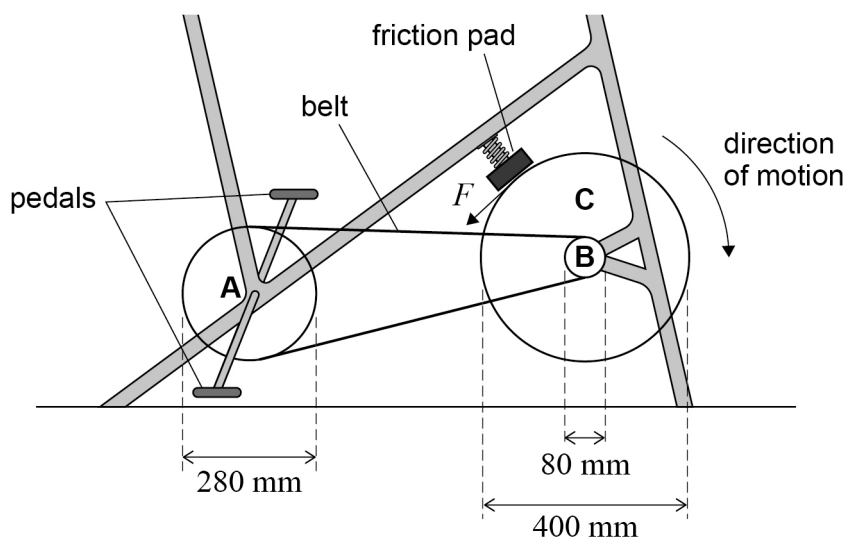
Question 5 continues on the next page



0 5 . 4

Figure 9 shows the drive system of the exercise bike, with the values for the diameters of **A**, **B** and **C** marked.

Figure 9



Determine the torque exerted by the belt on **A** during the first 2.3 s of **Figure 8** on page 16.

[4 marks]

torque = _____ N m



0 5 . 5

Describe the changes in the torque exerted by the rider during the time shown on the graph in **Figure 8** on page 16.

[2 marks]

11

Turn over for the next question

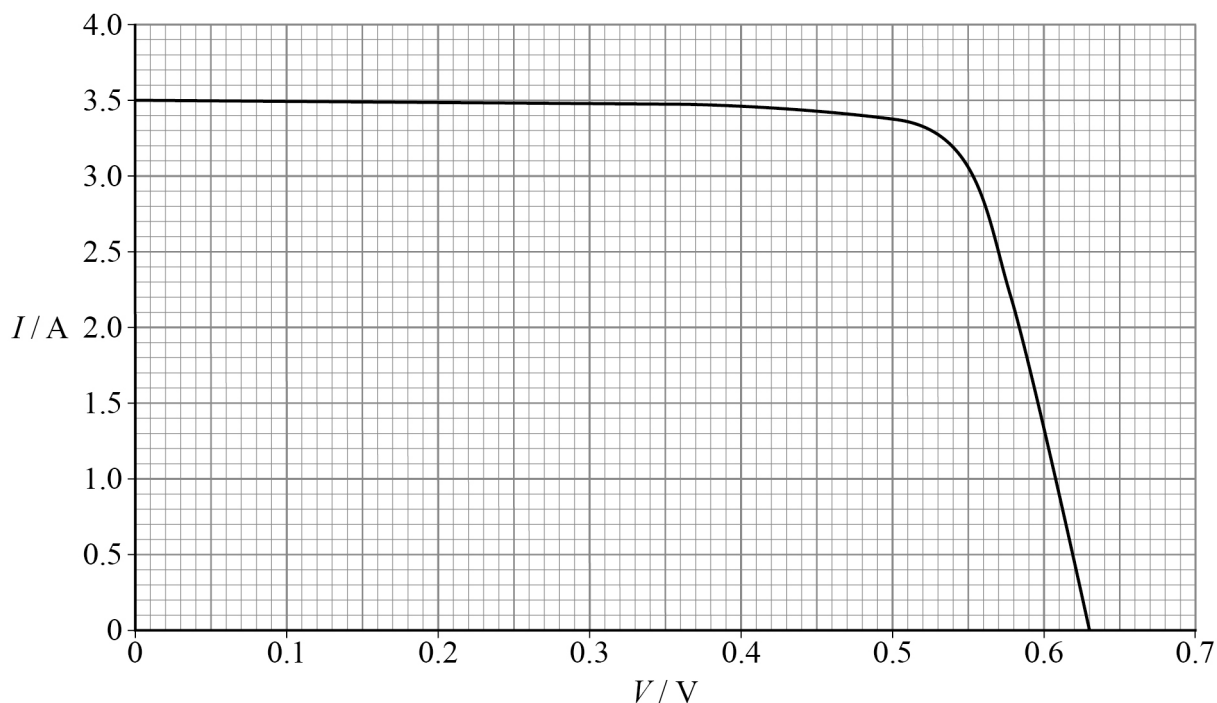
Turn over ►



0 6

Figure 10 shows the I – V characteristic for a solar cell.

Figure 10



The solar cell delivers its maximum power when the potential difference across its terminals is 0.54 V.

0 6 . 1

An engineer designs an arrangement of solar cells. Each solar cell has the I – V characteristic shown in **Figure 10**. The solar cells operate at their maximum power.

The arrangement must provide:

- a potential difference of at least 16 V
- a current of at least 10 A.

Determine how an arrangement of 120 solar cells can satisfy these conditions.

[2 marks]

space for calculation



0 6 . 2

The internal resistance r of each solar cell is given by:

$$r = \frac{0.63 \text{ V}}{3.5 \text{ A}}$$

Explain why it is appropriate to use 0.63 V and 3.5 A to calculate r .

[2 marks]

0.63 V _____

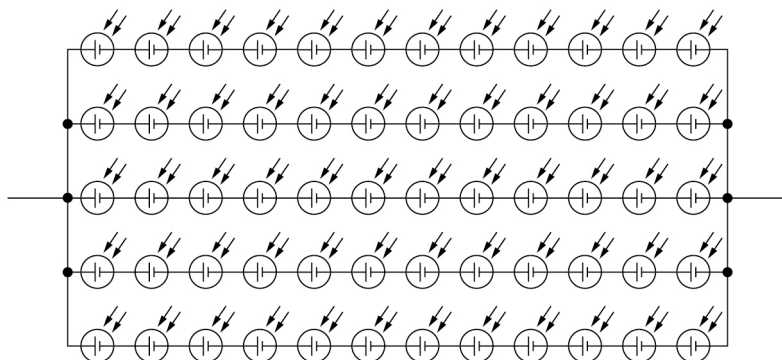
3.5 A _____

Question 6 continues on the next page



A different arrangement of solar cells has five parallel arms.
Each arm has 12 solar cells in series as shown in **Figure 11**.
Each cell has the I - V characteristic shown in **Figure 10** on page 20.

Figure 11



0 6 . 3

Determine the maximum power available from the arrangement in **Figure 11**.

[2 marks]

power = _____ W

0 6 . 4

Calculate the internal resistance of the arrangement in **Figure 11**.

[2 marks]

internal resistance = _____ Ω



0 6 . 5

Solar radiation is incident normal to the top of the Earth's atmosphere with an intensity of 1350 W m^{-2} .

The intensity of this solar radiation incident normal to the ground at the Earth's equator at midday is 1100 W m^{-2} .

Show that this reduction in intensity is **not** consistent with an inverse-square law.

radius of the Earth's orbit = $1.50 \times 10^{11} \text{ m}$

thickness of the Earth's atmosphere = 100 km

[2 marks]

10

END OF SECTION A

Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

0 7

A solid material is melted at constant temperature to form a liquid.

The mean potential energy of the molecules in the material is E_p .

The mean kinetic energy of the molecules in the material is E_k .

What happens to E_p and E_k during the melting process?

[1 mark]

	E_p	E_k	
A	stays the same	increases	☐
B	stays the same	stays the same	☐
C	increases	increases	☐
D	increases	stays the same	☐



0 8 The SI fundamental (base) unit of specific latent heat is:

[1 mark]

A J kg^{-1}

☐

B $\text{kg m}^2 \text{s}^{-2}$

☐

C $\text{m}^2 \text{s}^{-2}$

☐

D N m kg^{-1}

☐

0 9 A bar is made from two metal cylinders **R** and **S**, each with the same cross-sectional area.

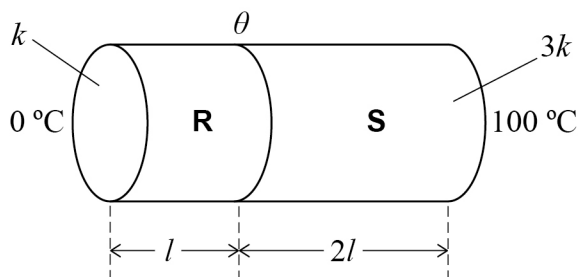
R has length l and thermal conductivity k .

S has length $2l$ and thermal conductivity $3k$.

The ends of the bar are maintained at temperatures of 0°C and 100°C .

The temperature at the junction between **R** and **S** is θ .

The rates of heat transfer through **R** and **S** are the same.



What is θ ?

[1 mark]

A 40°C

☐

B 50°C

☐

C 60°C

☐

D 67°C

☐

1 0

Two types of insulation block **P** and **Q** have the same dimensions.
The U-value of **P** is less than the U-value of **Q**.

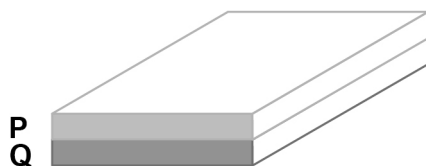
The diagrams show various combinations of these blocks.

The temperature difference between the top and bottom of each combination is the same.

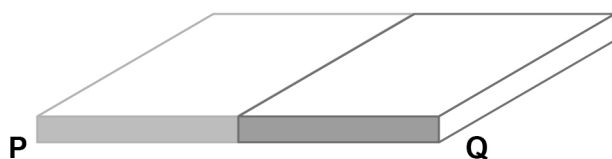
Which combination has the greatest rate of heat transfer?

[1 mark]

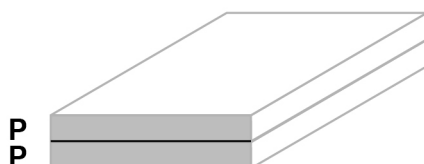
A



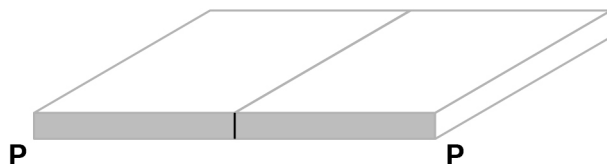
B



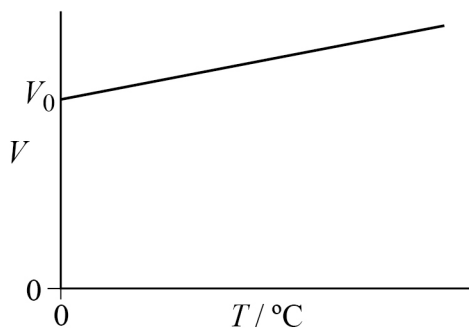
C



D



- 1 1** The graph shows the variation of volume V with temperature T for n mol of an ideal gas. The pressure p of the gas remains constant. The volume of the gas is V_0 at 0°C .



The gradient of the graph is:

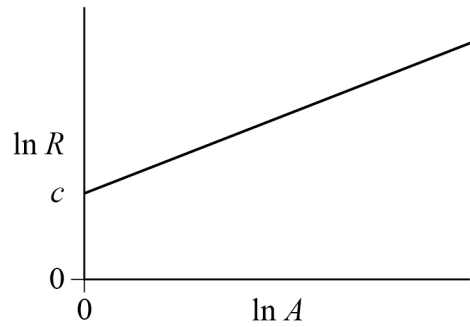
[1 mark]

- A** $\frac{p}{nR}$ ☐
- B** $\frac{nRT}{V_0}$ ☐
- C** $\frac{1}{273}$ ☐
- D** $\frac{V_0}{273}$ ☐

Turn over for the next question

Turn over ►

1 2 A graph is drawn of $\ln R$ against $\ln A$, where R is nuclear radius and A is nucleon number.



The radius of a hydrogen (${}^1_1\text{H}$) nucleus is R_0 .

Which row shows R_0 and the gradient of the graph?

[1 mark]

	R_0	Gradient of graph	
A	c	$\frac{1}{3}$	☐
B	e^c	3	☐
C	c	3	☐
D	e^c	$\frac{1}{3}$	☐

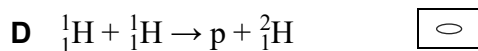
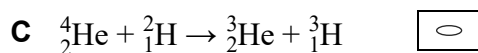
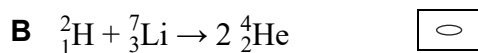
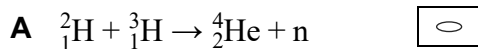


1 3

The table gives the nuclear masses of some nuclides.

Nuclide	Nuclear mass / u
${}^1_1\text{H}$	1.00783
${}^2_1\text{H}$	2.01410
${}^3_1\text{H}$	3.01605
${}^3_2\text{He}$	3.01603
${}^4_2\text{He}$	4.00260
${}^7_3\text{Li}$	7.01600
n	1.00866

Which reaction is possible without a net input of energy?

[1 mark]**Turn over for the next question****Turn over ►**

1 4

Which row gives desirable properties for the moderator, coolant and control rods in a thermal nuclear reactor?

[1 mark]

	Moderator	Coolant	Control rods	
A	poor neutron absorber	low specific heat capacity	good neutron absorber	☐
B	low nucleon number	high specific heat capacity	good neutron absorber	☐
C	good neutron absorber	high specific heat capacity	high nucleon number	☐
D	low nucleon number	low specific heat capacity	high nucleon number	☐

1 5

A solid sphere of uranium contains a mixture of uranium-235 and uranium-238 atoms. The uranium has a mass that is less than the critical mass.

Which change makes it possible for this mass of uranium to become critical?

[1 mark]

- A** increasing the ratio $\frac{\text{number of uranium-235 atoms}}{\text{number of uranium-238 atoms}}$ for the sphere ☐
- B** surrounding the sphere with a material that absorbs neutrons ☐
- C** changing the sphere into a cube of the same volume ☐
- D** increasing the temperature of the sphere ☐

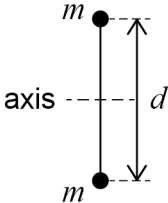
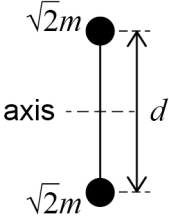
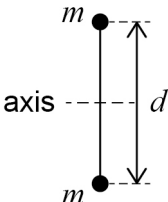
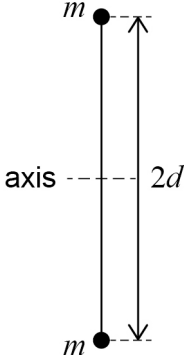
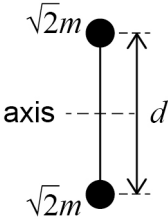
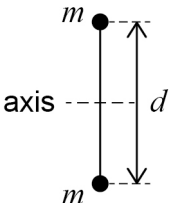
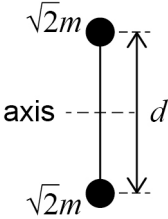
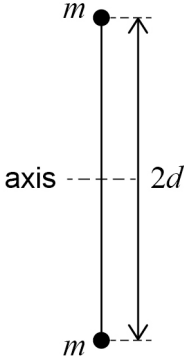
1 6

In the diagrams below, each object is a pair of point masses connected by a light rod. The axis of rotation is shown in each case.

In which row is:

moment of inertia of **Object 2** = $4 \times$ moment of inertia of **Object 1**?

[1 mark]

	Object 1	Object 2
A		
B		
C		
D		

1 7

Disc **P** rotates freely with an angular speed of 60.0 rad s^{-1} .

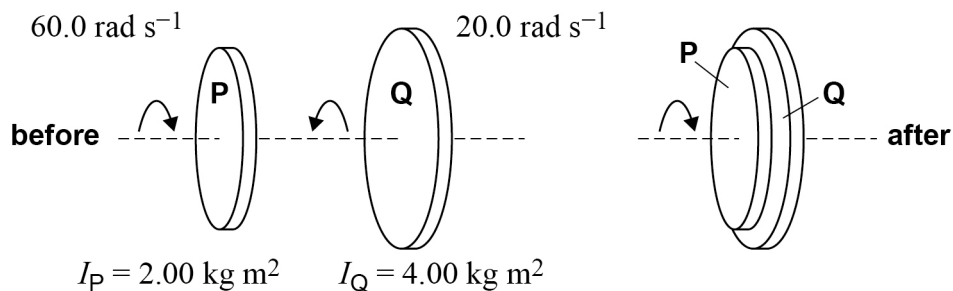
P has a moment of inertia I_P of 2.00 kg m^2 .

Disc **Q** rotates freely in the opposite direction to the rotation of **P**.

Q has an angular speed of 20.0 rad s^{-1} .

Q has a moment of inertia I_Q of 4.00 kg m^2 .

P and **Q** have the same axis of rotation.



P and **Q** now combine and rotate together. No external torque is involved.

What is the total rotational kinetic energy of **P** and **Q** after they combine?

[1 mark]

A 130 J

☐

B 270 J

☐

C 3300 J

☐

D 4400 J

☐
1 8

A student designs an investigation to discover whether a sample of a gas obeys Boyle's law.

Which **two** properties of the gas should be kept constant?

[1 mark]

A pressure and volume of gas

☐

B temperature and mass of gas

☐

C pressure and mass of gas

☐

D temperature and volume of gas

☐


1 9 Which is **not** equivalent to $\frac{1}{2}\pi r^2 \rho v^3$ for a wind turbine?

[1 mark]

- A** the maximum power that the turbine can extract from the wind passing through the turbine ☐
- B** the maximum power of the wind passing through the turbine ☐
- C** $\frac{p^2}{2m}$ where m is the mass and p is the momentum of the air passing through the turbine in one second ☐
- D** the kinetic energy of the air arriving at the area swept by the turbine's blades in one second ☐

2 0 A student plans to determine the specific heat capacity c of a metal block by an electrical method.

She plans to heat the block and measure its temperature rise.

The initial temperature of the block is T_1 .

The final temperature of the block is T_2 .

The constant temperature of the surroundings is T_s .

She wishes to reduce the effect of heat transfer between the block and the surroundings on her value of c .

Which relationship between T_1 , T_2 and T_s should she use?

[1 mark]

- A** $T_s < T_1 < T_2$ ☐
- B** $T_1 < T_2 < T_s$ ☐
- C** $T_1 < T_s < T_2$ ☐
- D** $T_s < T_2 < T_1$ ☐

Turn over for the next question

Turn over ►



2 1

Rain falling over an area of 280 km^2 is collected in a lake that is used by a hydroelectric power station.

The mean amount of rain falling in the area is 40 cm per year.

The turbine is 120 m below the water level in the dam. The water level of the lake stays constant.

The power station has an efficiency of 85%.

$$\text{density of water} = 1.0 \times 10^3 \text{ kg m}^{-3}$$

What is the mean output power available from the station?

[1 mark]

A 0.360 MW

☐

B 3.6 MW

☐

C 4.2 MW

☐

D 420 MW

☐

[illegible]

[illegible]

[illegible]

[illegible]

0	3
---	---

Experiments are done to determine the nuclear radius of an iron isotope **X** and of an iron isotope **Y**.

A nucleus of **X** contains 26 protons and 28 neutrons.

0	3
---	---

1

The radius of a nucleus of **X** is measured as 4.30×10^{-15} m using electron diffraction.

Determine the density of this nucleus.

[3 marks]

density = _____ kg m^{-3}

0	3
---	---

2

The radius of a nucleus of **Y** is measured as 4.40×10^{-15} m using electron diffraction.

Determine the number of neutrons in the nucleus of **Y**.

[3 marks]

number of neutrons = _____



The nuclear radius of **X** and the nuclear radius of **Y** are estimated using the closest approach of an alpha particle.

0 3 . 3

The upper limit for the radius of a nucleus of **X** is estimated to be 5.0×10^{-14} m.

Calculate the initial speed of an alpha particle that gives this estimate.
Assume that the nucleus remains at rest during the interaction.

[4 marks]

initial speed = _____ m s⁻¹

0 3 . 4

The upper limit for the radius of a nucleus of **Y** is estimated using an alpha particle with the same initial speed as in Question **03.3**.

Suggest a value for this estimate.
Explain your answer.

[1 mark]

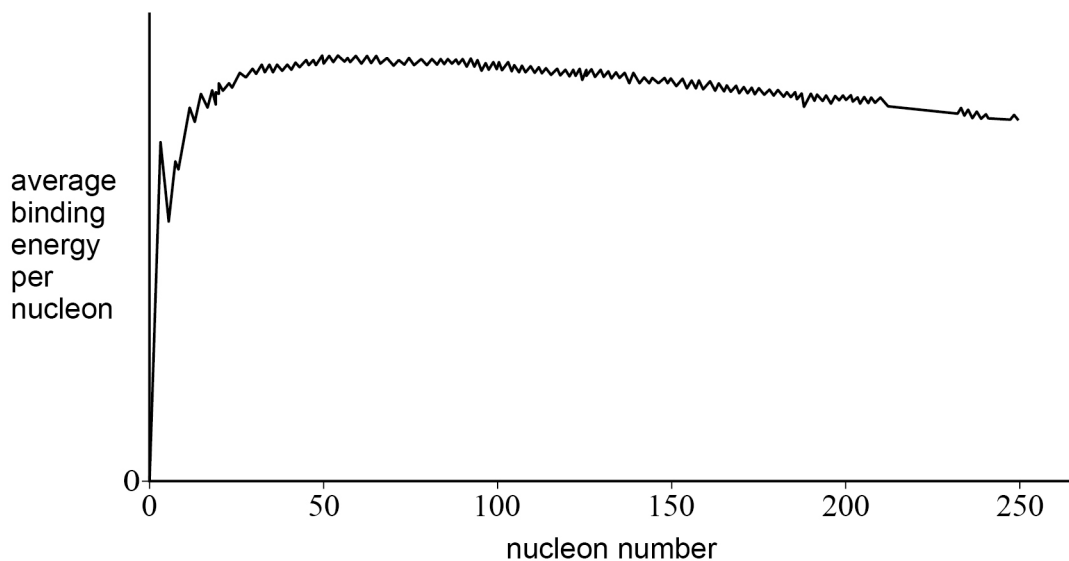
estimate of upper limit for radius = _____ m



Don't write

Figure 4 is a plot of average binding energy per nucleon against nucleon number for common nuclides.

Figure 4



0	4	.	1
---	---	---	---

Explain how:

- the fission of a nucleus and the fusion of nuclei can release energy
- the fission of a nucleus can release a greater amount of energy than the fusion of two nuclei.

Refer to relevant regions of **Figure 4** in your answer.

[4 marks]

[illegible]

0 4 . 2 The reaction below occurs in a nuclear reactor.

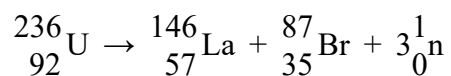


Table 3 gives data for the nuclides in this reaction.

Table 3

Nuclide	Binding energy per nucleon / MeV
${}_{92}^{236}\text{U}$	7.5865
${}_{57}^{146}\text{La}$	8.2400
${}_{35}^{87}\text{Br}$	8.6059

Calculate, in J, the energy released in the reaction.

[3 marks]

energy = _____ J

Question 4 continues on the next page



0 4 . 3

An iron $\left(\begin{smallmatrix} 56 \\ 26 \end{smallmatrix} \text{Fe} \right)$ nucleus has a mass of 55.935 u.

Calculate, in MeV, the binding energy of this nucleus.

[3 marks]

binding energy = _____ MeV

10



Do not write

0**8**

An efficient moderator in a thermal nuclear reactor must

[1 mark]

Do not write
outside the
box

A contain atoms that are good at absorbing neutrons.

☐

B be a good thermal conductor.

☐

C be a liquid at its operating temperature.

☐

D contain atoms with a small nucleon number.

☐

0**9**

What is the main purpose of lowering control rods into the core of a thermal nuclear reactor?

[1 mark]

A to decrease the number of thermal neutrons

☐

B to increase the number of thermal neutrons

☐

C to decrease the average kinetic energy of the thermal neutrons

☐

D to increase the average kinetic energy of the thermal neutrons

☐

A nuclear fission power station is designed to provide 45 MW of electrical power.

The energy output of the power station is 11% of the energy obtained by converting mass into energy.

What is the rate of conversion of mass into energy in the power station?

[1 mark]

A $4.5 \times 10^{-15} \text{ kg s}^{-1}$

☐

B $5.5 \times 10^{-11} \text{ kg s}^{-1}$

☐

C $5.0 \times 10^{-10} \text{ kg s}^{-1}$

☐

D $4.5 \times 10^{-9} \text{ kg s}^{-1}$

☐

Do not write

0 1

Table 1 gives the surface areas and U -values for the external parts of a house before and after additional insulation has been added.

Table 1

part of house	walls	doors	windows	roof
area / m ²	90	4	16	70
U -value with standard insulation / W m ⁻² K ⁻¹	1.0	2.4	5.0	0.6
U -value with additional insulation / W m ⁻² K ⁻¹	0.6	2.4	2.9	0.3

0 1 . 1

A house has standard insulation for the walls, windows, doors and roof. The temperature inside the house is 23 °C and the temperature outside is 34 °C.

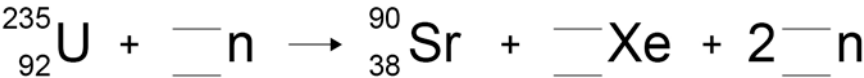
Calculate the total rate of heat transfer into the house.

[4 marks]

rate of heat transfer = _____ W

0 1 . 2

An example of induced fission is shown below.



Complete the equation.

[2 marks]

0 1 . 3

Each fission reaction transforms 200 MeV of energy into useful forms. A reactor produces 600 MW.

Calculate, in g s^{-1} , the rate of use of uranium-235

[3 marks]

rate of use of uranium-235 = _____ g s^{-1}

0 1 . 4

Determine which single part of the house should be insulated to give the best reduction in the rate of heat transfer. Use the data from **Table 1**. Justify your answer.

[3 marks]

0 1 . 5

Boron is used to make control rods for use in many fission reactors.

Explain the purpose of control rods and why boron is a suitable material for this use.

[3 marks]

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